

COUNTEREXAMPLES TO THE SCHIFFER AND POMPEIU CONJECTURES IN DIMENSIONS FOUR AND SIX

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ABSTRACT. For $n = 4$ and for $n = 6$ we construct a bounded convex domain $\Omega \subset \mathbb{R}^n$ which is not a ball, whose boundary is a real-analytic hypersurface diffeomorphic to S^{n-1} , and which carries a nonconstant Neumann eigenfunction of the Laplacian that is constant on $\partial\Omega$: the overdetermined problem $\Delta u + u = 0$ in Ω , $u = 1$ and $\partial_\nu u = 0$ on $\partial\Omega$ has a solution. By Green's identity the Fourier transform of the indicator function of Ω vanishes on the unit sphere, so Ω fails the Pompeiu property. This gives counterexamples to the Schiffer conjecture and to the Pompeiu conjecture in dimensions four and six. To our knowledge, they are the first counterexamples to either conjecture, in Euclidean space of any dimension, that are known to be convex. The four-dimensional domain is invariant under $O(3)$ acting on the last three coordinates and under reflection of the first one; the six-dimensional domain is invariant under $O(3) \times O(3)$ acting on $\mathbb{R}^3 \times \mathbb{R}^3$ and under the exchange of the two factors. For such domains the problem is equivalent to a planar Helmholtz problem in a meridian domain, with Cauchy data given on its boundary by the harmonic polynomial y , respectively $y_1 y_2$. The reduction is exact because conjugating the radial Laplacian of $\mathbb{R}^1$ or $\mathbb{R}^3$ by the square root of its radial Jacobian leaves no potential term. We treat the planar problems with the conformal fixed-disc method of Colbrook and Stepaniants, and prove in each case the existence of an exact solution near an explicit approximate one by a Newton–Kantorovich argument in a weighted space of disk-polynomial coefficients. The finite-dimensional parts of the arguments are checked in interval arithmetic, and all infinite tails are controlled by explicit analytic estimates.

CONTENTS

1. Introduction	2
1.1. The Pompeiu problem and Schiffer's conjecture	2
1.2. The planar counterexamples	3
1.3. Main results	4
1.4. Strategy of the proof	5
1.5. Other dimensions	6
Organisation	6
2. Reduction to a planar problem	6
3. The conformal fixed-disc formulation	8
4. Disk polynomials and coefficient spaces	9
4.1. Disk polynomials	9
4.2. The sine sector	11
5. The compatible inverse of the Laplacian	11
6. The cubic equation	12
6.1. Expansion at the centre	13
7. The contraction argument and the approximate inverse	13

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7.1.	A Newton–Kantorovich theorem	13
7.2.	Finite block and tail	14
7.3.	The approximate inverse	15
8.	The defect bounds	16
8.1.	The residual and the finite columns	16
8.2.	Tail g -columns	16
8.3.	Shape columns	16
9.	Certified bounds and the exact zero	19
10.	Reconstruction of the domain for $n = 4$	20
10.1.	Univalence, star-shapedness and the non-disc property	20
10.2.	Convexity	21
10.3.	Analytic regularity and continuation	21
10.4.	Proof of Theorem 1.1 for $n = 4$	22
11.	Dimension six	23
11.1.	Reduction to a planar problem	23
11.2.	Symmetry sector and conformal formulation	25
11.3.	The quartic equation	26
11.4.	Expansion at the centre	26
11.5.	The principal part of the linearisation	27
11.6.	Tail model and approximate inverse	28
11.7.	The defect bounds	29
11.8.	Certified bounds and the exact zero	31
11.9.	Reconstruction, geometry and convexity	32
11.10.	Proof of Theorem 1.1 for $n = 6$	34
12.	The Pompeiu property	34
13.	Computer-assisted verification	35
13.1.	Dimension four: what the script checks	35
13.2.	Dimension four: reproduction	35
13.3.	Dimension six	36
13.4.	Availability	37
13.5.	Consistency checks	38
14.	Remarks on other dimensions and on curved spaces	39
14.1.	Symmetry classes for which the reduction is exact	39
14.2.	A transcritical mechanism	39
14.3.	Dimensions three and five: numerical evidence	40
14.4.	Curved spaces	40
15.	Relation with rigidity results and other claims	41
	Acknowledgements	42
	References	42

1. INTRODUCTION

1.1. The Pompeiu problem and Schiffer’s conjecture. Let $n \geq 2$. A bounded domain $\Omega \subset \mathbb{R}^n$ has the *Pompeiu property* if the only continuous function $f: \mathbb{R}^n \rightarrow \mathbb{C}$ such that

$$\int_{\sigma(\Omega)} f(x) dx = 0 \quad \text{for every rigid motion } \sigma \text{ of } \mathbb{R}^n$$

is $f \equiv 0$. The question of which domains have this property goes back to Pompeiu [30–32]. Balls do not have it: if $J_{n/2}(kR) = 0$ for some $k > 0$, then the Fourier transform of $\mathbf{1}_{B_R}$ vanishes on the sphere $\{|\xi| = k\}$, and $f(x) = e^{ikx_1}$ integrates to zero over every ball of radius R . Brown, Schreiber and Taylor [5] characterised the failure of the Pompeiu property through the zero set of the Fourier–Laplace transform of the indicator function; in particular, a bounded domain fails the property whenever $\widehat{\mathbf{1}_\Omega}$ vanishes identically on a sphere $\{|\xi| = k\}$ with $k > 0$ (this direction is elementary, see Section 12). The *Pompeiu conjecture* asserts that a bounded domain with Lipschitz boundary homeomorphic to S^{n-1} fails the Pompeiu property only if it is a ball. Some topological assumption is necessary: for every $k > 0$ the function $R \mapsto R^{n/2} J_{n/2}(kR)$ takes equal values at suitable distinct radii $R_1 < R_2$, so the spherical shell $\{R_1 < |x| < R_2\}$ fails the Pompeiu property, and such shells are simply connected when $n \geq 3$.

Conjecture 1 (Schiffer). *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with smooth boundary homeomorphic to S^{n-1} . If, for some $\mu > 0$, there is a nonconstant solution of*

$$(1.1) \quad \Delta u + \mu u = 0 \quad \text{in } \Omega, \quad u = \text{const} \quad \text{and} \quad \partial_\nu u = 0 \quad \text{on } \partial\Omega,$$

then Ω is a ball.

On a ball B_R , the radial Neumann eigenfunctions $u = r^{1-n/2} J_{n/2-1}(\sqrt{\mu} r)$ solve (1.1) precisely when $J_{n/2}(\sqrt{\mu} R) = 0$. Formulations of the conjecture in the literature differ in their regularity and topological hypotheses (smooth boundary, simply connected domain, connected boundary); the domains constructed below satisfy all of them. Williams [38] showed that, for bounded domains with Lipschitz boundary homeomorphic to a sphere, the failure of the Pompeiu property is equivalent to the solvability of (1.1) for some $\mu > 0$, building on [5]; the direction from (1.1) to the failure of the Pompeiu property is a consequence of Green’s identity. In the planar case Schiffer’s problem appears as Problem 80 in Yau’s list [39]. We refer to Zalcman’s survey [40] for the classical bibliography.

A large literature proves the conjectured rigidity under additional hypotheses. Berenstein [3] and Berenstein–Yang [4] showed that a domain with infinitely many eigenvalues μ for which (1.1) is solvable is a ball. Kobayashi [27], Agranovsky [1] and Canuto [6] proved rigidity and stability statements for perturbations of balls. Further results under geometric or spectral hypotheses include [2, 12–15, 26, 29]. We compare our examples with these results in Section 15.

1.2. The planar counterexamples. In August 2026 the planar conjectures were disproved independently by two groups. Colbrook and Stepaniants [11] constructed a single explicit ten-fold symmetric noncircular domain $\Omega \subset \mathbb{R}^2$ with real-analytic boundary for which (1.1) holds with $\sqrt{\mu} \in (31.967007261, 31.967007293)$. Their proof pulls the problem back to the unit disc by a conformal map, which turns it into a cubic equation on a space of disk-polynomial coefficients, and establishes the existence of an exact solution near an explicit approximation by a computer-assisted contraction argument. Cao-Labora and de Dios Pont [7] constructed infinitely many N -fold symmetric planar counterexamples, for N sufficiently large, by a bifurcation argument for a relaxed problem in which N is a real parameter, with a branch length that is uniform in N . Colbrook, Sadeghi and Stepaniants [10] adapted the conformal method to disprove the planar Berenstein conjecture. Earlier constructions of Schiffer-type domains in related settings include domains in the flat cylinder $\mathbb{R}^N \times \mathbb{R}/2\pi\mathbb{Z}$ and on S^2 [19], contractible domains on the half-sphere [8], doubly connected planar domains on whose two boundary components the eigenfunction takes different constants [18], and noncircular planar solutions of the related problem with nonzero constant Neumann data [37]. To the best of our knowledge, no counterexample to either conjecture was previously known for bounded domains in $\mathbb{R}^n$, $n \geq 3$, with boundary homeomorphic to a sphere.

1.3. Main results. Write $x = (x_1, x') \in \mathbb{R} \times \mathbb{R}^3$ for points of $\mathbb{R}^4$ and $x = (x', x'') \in \mathbb{R}^3 \times \mathbb{R}^3$ for points of $\mathbb{R}^6$.

Theorem 1.1. *Let $n \in \{4, 6\}$. There exist a bounded domain $\Omega \subset \mathbb{R}^n$ and a nonconstant real-valued function u , real analytic on an open neighbourhood of $\overline{\Omega}$, such that*

$$(1.2) \quad \Delta u + u = 0 \text{ in } \Omega, \quad u = 1 \text{ and } \nabla u = 0 \text{ on } \partial\Omega.$$

Moreover:

- (i) Ω is not a ball;
- (ii) Ω is strictly star-shaped with respect to the origin, and $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^{n-1} ;
- (iii) for $n = 4$, Ω is invariant under the action of $O(3)$ on x' and under $x_1 \mapsto -x_1$; for $n = 6$, Ω is invariant under the action of $O(3) \times O(3)$ on (x', x'') and under the exchange $(x', x'') \mapsto (x'', x')$.

In particular, Conjecture 1 fails in $\mathbb{R}^4$ and in $\mathbb{R}^6$.

Rescaling Ω and u gives, for every $\mu > 0$, a counterexample with eigenvalue μ in (1.1).

Corollary 1.2 (Pompeiu). *Let $n \in \{4, 6\}$ and let Ω be the domain of Theorem 1.1. The Fourier transform of $\mathbf{1}_\Omega$ vanishes on the unit sphere of $\mathbb{R}^n$. Consequently $\int_{\sigma(\Omega)} \cos x_1 dx = 0$ for every rigid motion σ of $\mathbb{R}^n$, the domain Ω fails the Pompeiu property, and the Pompeiu conjecture fails in $\mathbb{R}^4$ and in $\mathbb{R}^6$.*

Corollary 1.3 (Convexity). *For $n = 4$ and for $n = 6$, the domain Ω of Theorem 1.1 is convex.*

In particular, the Schiffer and Pompeiu conjectures fail in $\mathbb{R}^4$ and in $\mathbb{R}^6$ even within the class of bounded convex domains with real-analytic boundary. To our knowledge, the domains of Theorem 1.1 are the first counterexamples to either conjecture, in Euclidean space of any dimension, that are known to be convex. Neither [11] nor [7] makes a convexity claim; the domains of [7] are asserted to be star-shaped. The domain of [11] is not convex: the certified enclosure of its conformal map φ gives $\operatorname{Re}(1 + z\varphi''(z)/\varphi'(z)) \in (-2.3156, -2.3155)$ at the ten boundary points midway between consecutive lobes, so the boundary curvature is negative there. Nor is the 28-fold example displayed in [7, Figure 1], whose boundary is visibly indented between consecutive lobes. Heuristically, both planar constructions perturb a disc by a mode of high angular frequency N , and a relative amplitude ε changes the boundary curvature by a relative amount of order εN^2 ; our domains instead deviate from a ball mainly by a single zonal harmonic of low degree and small relative amplitude (Remark 1.4). Both of our domains are moreover centrally symmetric, since $-\operatorname{id}$ lies in their symmetry groups. The planar rigidity results [13, 29] are stated for (uniformly) convex domains; they have no counterpart in higher dimensions, and Section 15 discusses what our examples imply for one. As far as we know, it remains open whether a convex counterexample exists in the plane.

Remark 1.4 (The domains). Both domains are obtained from a planar meridian domain $D = \psi(\mathbb{D}) \subset \mathbb{R}^2 \cong \mathbb{C}$, the image of the unit disc under a map ψ with real Taylor coefficients.

Dimension four. Here $\Omega = \{(x_1, x') : (x_1, |x'|) \in D\}$ and $\psi(w) = \sum_{j \text{ odd}} c_j w^j$. The coefficients satisfy

$$\sum_{j \text{ odd}} (c_1^\circ)^2 (j+1) \left(\frac{21}{20}\right)^j |c_j - c_j^\circ| \leq \frac{1}{50000},$$

where $c_1^\circ, c_3^\circ, \dots, c_{41}^\circ$ are explicit dyadic rationals listed in the verification script (Section 13) and $c_j^\circ = 0$ for $j > 41$; one has $c_1^\circ \approx 14.840043$ and $c_3^\circ \approx 0.0912$, $c_5^\circ \approx 0.0942$, $c_7^\circ \approx 0.0964$. For orientation we record some numerical observations about the centre ψ° , which are not

used in the proofs. The boundary of D is a radial graph whose radius lies between 14.7213 and 15.1343. The ball of $\mathbb{R}^4$ that carries a radial solution of (1.2) with the same eigenvalue and closest radius has radius $j_{2,4} \approx 14.79595$, the fourth positive zero of J_2 ; thus the relative deviation of $\partial\Omega$ from that sphere lies between -0.5% and $+2.3\%$. Expanding the boundary radius of Ω in zonal spherical harmonics on S^3 , the dominant nonconstant term is the zonal harmonic of degree six, with relative amplitude about 2.2×10^{-2} at the poles.

Dimension six. Here $\Omega = \{(x', x'') : (|x'|, |x''|) \in D\}$ and $\psi(w) = \sum_{j \equiv 1 \pmod{4}} c_j w^j$. The coefficients satisfy

$$\sum_{j \equiv 1 \pmod{4}} \frac{(c_1^\circ)^3}{2} (j+2) \left(\frac{11}{10}\right)^{j+1} |c_j - c_j^\circ| \leq \frac{1}{5000},$$

where $c_1^\circ, c_5^\circ, \dots, c_{57}^\circ$ are explicit dyadic rationals listed in the ancillary data file `center_r6.json` (Section 13.3) and $c_j^\circ = 0$ for $j > 57$; one has $c_1^\circ \approx 28.864751$, $c_5^\circ \approx -0.0852970$ and $c_9^\circ \approx -0.0837734$, while $|c_j^\circ| < 4 \cdot 10^{-4}$ for $j \geq 13$. Numerically, the boundary of D is a radial graph whose radius lies between 28.6958 and 28.9596, and the ball of $\mathbb{R}^6$ with the same volume as Ω has radius 28.908339. The ball of $\mathbb{R}^6$ that carries a radial solution of (1.2) with the same eigenvalue and closest radius has radius $j_{3,8} \approx 28.908351$, the eighth positive zero of J_3 ; the relative deviation of $\partial\Omega$ from that sphere lies between -0.74% and $+0.18\%$, so that the radius of $\partial\Omega$ varies by about 0.92% . Expanding the boundary radius in the $O(3) \times O(3)$ -invariant spherical harmonics on S^5 , which are the functions $U_k(|\omega'|^2 - |\omega''|^2)$ of degree $2k$ (Section 14.2), the dominant nonconstant term has degree eight, with relative amplitude about 7.2×10^{-3} .

1.4. Strategy of the proof. The proofs for $n = 4$ and $n = 6$ have the same structure. We give the details for $n = 4$ in Sections 2–10, and then, in Section 11, the proof for $n = 6$, which uses most lemmas of the four-dimensional proof verbatim. For $n = 4$ the proof has three parts.

Reduction to a planar problem (Section 2). For $u(x) = f(x_1, |x'|)$, the function $F(x_1, y) = yf(x_1, y)$ satisfies

$$(1.3) \quad (\partial_{x_1}^2 + \partial_y^2)F = y(\Delta_{\mathbb{R}^4}u)(x_1, y, 0, 0),$$

because the radial part of the Laplacian of $\mathbb{R}^3$ is $y^{-1}\partial_y^2(y \cdot)$. Hence $O(3) \times \mathbb{Z}_2$ -invariant solutions of (1.2) correspond to solutions of the planar problem

$$(1.4) \quad \Delta F + F = 0 \text{ in } D, \quad F = y \text{ and } \nabla F = e_y \text{ on } \partial D,$$

with F odd in y and D symmetric under both coordinate reflections.

Conformal fixed-disc formulation and contraction (Sections 3–9). Following [11] we write $D = \psi(\mathbb{D})$ and $V = F \circ \psi$. Then $\Delta V + |\psi'|^2 V = 0$ in $\mathbb{D}$, and $V - \text{Im } \psi$ has vanishing value and normal derivative on $\partial\mathbb{D}$. We write $V = Kg + \text{Im } \psi$, where K is an explicit inverse of the Laplacian with vanishing Cauchy data on its natural range, and obtain the cubic equation

$$(1.5) \quad \mathcal{F}(g, \psi) = g + |\psi'|^2 (Kg + \text{Im } \psi) = 0.$$

Compared with the equation $g + |p|^2(1 + Kg) = 0$ of [11], the constant boundary datum is replaced by $\text{Im } \psi$, so that the unknown map enters the data as well; the eigenvalue is absorbed into the scale of ψ ; the symmetry class consists of odd maps with real coefficients and the sine sector of odd angular frequencies; and the part of the linearisation acting on high shape modes is an upper bidiagonal operator that we invert explicitly. We prove that (1.5) has an exact zero in an explicit ball around a numerically computed centre, using a Newton–Kantorovich (radii polynomial) argument in weighted ℓ^1 spaces of disk-polynomial coefficients. The approximate inverse consists of a 525×525 matrix, inverted and checked in ball arithmetic, together with an exact tail inverse. All infinitely many columns of the defect operator are covered: finitely many

are evaluated without truncation in interval arithmetic, and the remaining ones are bounded by analytic estimates proved in Section 8.

Reconstruction (Section 10). From the zero of (1.5) we obtain a univalent map ψ , a strictly star-shaped and convex meridian domain D , a solution of (1.4) that extends analytically across ∂D , and finally, by rotation, the domain Ω and the function u of Theorem 1.1.

Dimension six (Section 11). For $u(x) = f(|x'|, |x''|)$ on $\mathbb{R}^6 = \mathbb{R}^3 \times \mathbb{R}^3$, the function $F(y_1, y_2) = y_1 y_2 f(y_1, y_2)$ satisfies $\Delta_{\mathbb{R}^2} F = y_1 y_2 \Delta_{\mathbb{R}^6} u$, since the radial conjugation above can be applied in each factor. The planar problem becomes

$$(1.6) \quad \Delta F + F = 0 \text{ in } D, \quad F = y_1 y_2 \text{ and } \nabla F = (y_2, y_1) \text{ on } \partial D,$$

with F odd in y_1 and in y_2 , even under $(y_1, y_2) \mapsto (y_2, y_1)$, and D invariant under the symmetry group of the square. In the conformal formulation the Dirichlet datum $\text{Im} \psi$ is replaced by $\text{Im}(\psi^2)/2$, so that the equation becomes quartic, the maps are $\psi(w) = \sum_{j \equiv 1 \pmod{4}} c_j w^j$, and the unknowns lie in the sine sector of angular frequencies $\equiv 2 \pmod{4}$. The high shape modes are now governed by an explicit operator whose harmonic part is a difference operator of step four, which we again invert exactly. The remaining changes concern the constants: the approximate inverse is a 465×465 matrix, and the tail estimates are adapted to the new weights.

The four-dimensional finite computations are performed by a single self-contained script using ball arithmetic [20, 25]; it runs in about three minutes on one core. The six-dimensional ones are split into five stage scripts, whose interval receipts are combined by a final verification script; the complete reproduction also takes about three minutes on one core. Section 13 describes what the scripts check and how to reproduce the computations.

1.5. Other dimensions. Among the splittings $\mathbb{R}^n = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ with symmetry group $O(d_1) \times O(d_2)$, the reduction to a planar problem for the Laplacian, without a potential term, is available exactly when $d_1, d_2 \in \{1, 3\}$, that is, for $n \in \{2, 4, 6\}$ (Remark 2.5). Our proofs are therefore specific to dimensions four and six. In Section 14 we record numerical evidence, which we do not claim as proof, for analogous non-ball solutions of (1.1) in $\mathbb{R}^3$ and $\mathbb{R}^5$, discuss the bifurcation mechanism that suggested where to look, and point out other cohomogeneity-two symmetry classes, in dimensions 8, 10 and 14, to which the planar reduction extends (Remark 14.1); we have not pursued them. We do not know whether counterexamples exist in dimensions three and five, or in any dimension $n \geq 7$.

Organisation. Section 2 contains the reduction to the plane for $n = 4$ and Section 3 the conformal formulation. Section 4 collects the needed facts on disk polynomials and Section 5 studies the operator K . The cubic map and its derivatives are described in Section 6, the approximate inverse in Section 7, and the defect bounds, including all tail estimates, in Section 8. Section 9 states the certified bounds and the existence of the zero, and Section 10 proves Theorem 1.1 and Corollary 1.3 for $n = 4$. Section 11 contains the proof for $n = 6$. Section 12 proves Corollary 1.2, and Section 13 describes the computer-assisted parts. Sections 14 and 15 contain remarks on other dimensions and curved spaces, and on the relation with known rigidity results.

2. REDUCTION TO A PLANAR PROBLEM

Sections 2–10 contain the proof for $n = 4$; the proof for $n = 6$ is given in Section 11. Write points of the meridian plane as $(x_1, y) \in \mathbb{R}^2$, identified with $z = x_1 + iy \in \mathbb{C}$, and let $\Pi: \mathbb{R}^4 \rightarrow \mathbb{R}^2$, $\Pi(x_1, x') = (x_1, |x'|)$. The map Π is continuous and its image is the closed half-plane $\{y \geq 0\}$.

Lemma 2.1. *Let f be a C^2 function on an open subset of $\{y > 0\}$ and $F(x_1, y) = yf(x_1, y)$. Then*

$$F_{x_1x_1} + F_{yy} = y \left(f_{x_1x_1} + f_{yy} + \frac{2}{y} f_y \right).$$

The expression in parentheses, evaluated at $(x_1, |x'|)$, is the Laplacian of the function $x \mapsto f(x_1, |x'|)$ at every point with $x' \neq 0$.

Proof. $(yf)_{yy} = yf_{yy} + 2f_y$. The second statement is the formula $\partial_y^2 + \frac{2}{y}\partial_y$ for the Laplacian of $\mathbb{R}^3$ acting on radial functions. $\square$

Lemma 2.2. *Let $N \subset \mathbb{R}^2$ be open and symmetric under $(x_1, y) \mapsto (x_1, -y)$, and let F be real analytic on N and odd in y . Then there is a real-analytic function G , defined on an open set containing $\{(x_1, y^2) : (x_1, y) \in N\}$, such that $F(x_1, y) = yG(x_1, y^2)$ on N . Consequently $x \mapsto G(x_1, |x'|^2)$ is real analytic on the open set $\Pi^{-1}(N)$, and it equals $F(x_1, |x'|)/|x'|$ where $x' \neq 0$.*

Proof. For $t > 0$ with $(x_1, \sqrt{t}) \in N$ put $G(x_1, t) = F(x_1, \sqrt{t})/\sqrt{t}$; this is real analytic. Let $(a, 0) \in N$. For some $\delta > 0$, $F(x_1, y) = \sum_{i,k \geq 0} c_{ik}(x_1 - a)^i y^k$ converges absolutely for $|x_1 - a| < \delta$, $|y| < \delta$, and $c_{ik} = 0$ for even k by oddness. Then $G(x_1, t) = \sum_{i,j} c_{i,2j+1}(x_1 - a)^i t^j$ converges for $|x_1 - a| < \delta$, $|t| < \delta^2$ and agrees with the previous definition where $0 < t < \delta^2$. Two such local definitions agree on their overlap, which is a connected product of intervals containing points with $t > 0$. This defines G . The last assertion follows because $x \mapsto (x_1, |x'|^2)$ is polynomial. $\square$

Proposition 2.3 (From the plane to $\mathbb{R}^4$). *Let $D \subset \mathbb{R}^2$ be a bounded domain, symmetric under $y \mapsto -y$, whose boundary is a Jordan curve meeting the axis $\{y = 0\}$ in exactly two points. Let F be real analytic on an open set $N \supset \bar{D}$ symmetric under $y \mapsto -y$, odd in y , and such that*

$$(2.1) \quad \Delta F + F = 0 \text{ in } D, \quad F = y \text{ and } \nabla F = e_y \text{ on } \partial D.$$

Put $\Omega = \Pi^{-1}(D)$ and let u be the function $x \mapsto G(x_1, |x'|^2)$ of Lemma 2.2. Then Ω is a bounded open set, u is real analytic on the neighbourhood $\Pi^{-1}(N)$ of $\bar{\Omega}$, u is nonconstant, and (1.2) holds.

Proof. Ω is open and bounded because Π is continuous and proper. By continuity $\bar{\Omega} \subset \Pi^{-1}(\bar{D}) \subset \Pi^{-1}(N)$, and $\partial\Omega \subset \Pi^{-1}(\partial D)$, since a point $x \in \bar{\Omega}$ with $\Pi(x) \in D$ lies in the open set Ω .

Off the axis $\{x' = 0\}$, $u(x) = f(x_1, |x'|)$ with $f = F/y$, so Lemma 2.1 gives $|x'|(\Delta u + u)(x) = (\Delta F + F)(\Pi(x)) = 0$ in $\Omega \setminus \{x' = 0\}$. By continuity of $\Delta u + u$ the equation holds in Ω .

Let $x \in \partial\Omega$ with $y = |x'| > 0$. Then $\Pi(x) \in \partial D$, so $f = F/y = 1$ there. Since $\nabla F = e_y$ on ∂D , we have $f_{x_1} = F_{x_1}/y = 0$ and $f_y = (F_y - f)/y = 0$, hence $\nabla u(x) = (f_{x_1}, f_y x'/|x'|) = 0$. The set $\partial\Omega \cap \{x' \neq 0\} = \Pi^{-1}(\partial D \cap \{y > 0\})$ is dense in $\partial\Omega$, since ∂D meets the axis in only two points and each of them is a limit of points of $\partial D \cap \{y > 0\}$. By continuity $u = 1$ and $\nabla u = 0$ on all of $\partial\Omega$. Finally, a constant solution of $\Delta u + u = 0$ vanishes identically, which contradicts $u = 1$ on $\partial\Omega$. $\square$

The reduction is reversible in the analytic class, which explains why nothing is lost by working in the plane.

Proposition 2.4 (From $\mathbb{R}^4$ to the plane). *Let $\Omega \subset \mathbb{R}^4$ be a bounded domain invariant under $O(3)$ acting on x' , and let u be real analytic on a neighbourhood of $\bar{\Omega}$ and solve (1.2). Put $D = \{(x_1, y) : (x_1, |y|, 0, 0) \in \Omega\}$ and $\tilde{u}(x) = \int_{O(3)} u(x_1, Qx') dQ$ (Haar probability measure).*

Then $\tilde{u}$ also solves (1.2), it has the form $\tilde{u}(x) = f(x_1, |x'|)$ with f real analytic and even in y , and $F = yf$ solves (2.1).

Proof. The Laplacian commutes with rotations, and the boundary conditions are rotation invariant; hence $\tilde{u}$ solves (1.2). The function $f(x_1, y) = \tilde{u}(x_1, y, 0, 0)$ is real analytic and even in y (use the rotation $x' \mapsto -x'$), and $\tilde{u}(x) = f(x_1, |x'|)$ by invariance. Lemma 2.1 gives $\Delta F + F = 0$ in $D \cap \{y \neq 0\}$, and by analyticity in D . On $\partial D \cap \{y > 0\}$ we have $f = 1$ and $\nabla f = 0$, hence $F = y$ and $\nabla F = fe_y + y\nabla f = e_y$; continuity extends this to the points of ∂D on the axis. $\square$

Remark 2.5 (Why dimensions four and six). For a function on $\mathbb{R}^d$ that depends only on $y = |\xi|$, $\xi \in \mathbb{R}^d$, the Laplacian is $\partial_y^2 + (d-1)y^{-1}\partial_y$, and for $F = y^{(d-1)/2}f$ one has

$$\partial_y^2 F = y^{(d-1)/2} \left(f_{yy} + \frac{d-1}{y} f_y \right) + \frac{(d-1)(d-3)}{4} \frac{F}{y^2}.$$

Let $\mathbb{R}^n = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ and let $u(x) = f(y_1, y_2)$ be invariant under $O(d_1) \times O(d_2)$, where y_i is the modulus of the i -th component of x (for $d_i = 1$ one may use the coordinate itself). Then $F = y_1^{(d_1-1)/2} y_2^{(d_2-1)/2} f$ satisfies

$$F_{y_1 y_1} + F_{y_2 y_2} = y_1^{(d_1-1)/2} y_2^{(d_2-1)/2} \Delta u + \left(\frac{(d_1-1)(d_1-3)}{4y_1^2} + \frac{(d_2-1)(d_2-3)}{4y_2^2} \right) F.$$

The inverse-square potential vanishes exactly when $d_1, d_2 \in \{1, 3\}$, that is, for $n = d_1 + d_2 \in \{2, 4, 6\}$. In these cases the weight $\varphi = y_1^{(d_1-1)/2} y_2^{(d_2-1)/2}$ is one of the harmonic polynomials $1, y, y_1 y_2$, and the boundary conditions $u = 1, \nabla u = 0$ become $F = \varphi, \nabla F = \nabla \varphi$. For $n = 4$ ($d_1 = 1, d_2 = 3$) this is Lemma 2.1, and for $n = 6$ ($d_1 = d_2 = 3$) it is Lemma 11.1 below. The conformal method uses in an essential way that the planar operator is the Laplacian, whose conformal pull-back is a multiple of the Laplacian. For $n = 3$ and $n = 5$ every splitting into two blocks contains a block of dimension two or four, and the meridian problem contains a first-order drift term, or equivalently an inverse-square potential; a different fixed-domain formulation would be needed. Symmetry classes of other types are discussed in Remark 14.1.

Let $x_1 + iy = re^{i\phi}$ be planar polar coordinates, and let U_ℓ denote the Chebyshev polynomial of the second kind, so that $U_\ell(\cos \phi)$ is the zonal spherical harmonic of degree ℓ on S^3 . Under $u = F/y$, the planar Helmholtz solution $F = J_m(r) \sin m\phi$, with m odd, corresponds to the four-dimensional Helmholtz solution

$$u = r^{-1} J_m(r) U_{m-1}(\cos \phi).$$

For instance, $F = J_1(r) \sin \phi$ gives the radial solution $r^{-1} J_1(r)$, whose derivative is $-r^{-1} J_2(r)$; this is why the Schiffer balls of $\mathbb{R}^4$ at eigenvalue one have radii $j_{2,k}$, the positive zeros of J_2 .

3. THE CONFORMAL FIXED-DISC FORMULATION

Let $\mathbb{D} = \{w \in \mathbb{C} : |w| < 1\}$, $w = re^{i\theta}$. The following lemma pulls (2.1) back to the disc; it is the analogue of the reformulation in [11].

Lemma 3.1. *Let ψ be holomorphic and injective on a neighbourhood of $\overline{\mathbb{D}}$, with $\psi' \neq 0$ there, and let $D = \psi(\mathbb{D})$. Let $V \in C^1(\overline{\mathbb{D}})$ be real valued and satisfy*

$$(3.1) \quad \Delta V + |\psi'|^2 V = 0 \quad \text{in } \mathcal{D}'(\mathbb{D}), \quad V - \operatorname{Im} \psi = 0 \quad \text{and} \quad \nabla(V - \operatorname{Im} \psi) = 0 \quad \text{on } \partial \mathbb{D}.$$

Then $F = V \circ \psi^{-1} \in C^1(\overline{D})$ satisfies $\int_D (\nabla F \cdot \nabla \varphi - F \varphi) dA = 0$ for all $\varphi \in C_c^\infty(D)$, and $F = y, \nabla F = e_y$ on ∂D . If moreover ψ is odd with real Taylor coefficients, $V(\bar{w}) = -V(w)$

and $V(-\bar{w}) = V(w)$, then D is symmetric under both coordinate reflections, F is odd in y and even in x_1 .

Proof. For $\varphi \in C_c^\infty(D)$, the function $\varphi \circ \psi$ belongs to $C_c^\infty(\mathbb{D})$. Conformal invariance of the Dirichlet integral and the Jacobian $|\psi'|^2$ give

$$\int_D \nabla F \cdot \nabla \varphi \, dA = \int_{\mathbb{D}} \nabla V \cdot \nabla (\varphi \circ \psi) \, dA = \int_{\mathbb{D}} |\psi'|^2 V(\varphi \circ \psi) \, dA = \int_D F \varphi \, dA,$$

where the middle equality is the distributional equation for $V \in C^1$. The function $V - \text{Im } \psi = (F - y) \circ \psi$ has zero value and gradient on $\partial\mathbb{D}$; since the Jacobian matrix of ψ is invertible on $\partial\mathbb{D}$, the chain rule gives $F - y = 0$ and $\nabla(F - y) = 0$ on ∂D . If ψ is odd with real coefficients, then $\psi(\bar{w}) = \overline{\psi(w)}$ and $\psi(-\bar{w}) = -\overline{\psi(w)}$, which gives the symmetries of D and, with the stated parities of V , those of F . $\square$

We shall look for V of the form $V = Kg + \text{Im } \psi$, where g is an unknown function on $\mathbb{D}$ and K is an inverse of the Laplacian with zero Cauchy data on $\partial\mathbb{D}$ (Section 5). Since $\text{Im } \psi$ is harmonic, (3.1) becomes $g + |\psi'|^2(Kg + \text{Im } \psi) = 0$, which is (1.5). We normalise the eigenvalue to one; the size of D is then an unknown, encoded in the coefficient c_1 of ψ . The symmetry class removes the degeneracies of the problem: the scaling is fixed by the eigenvalue, translations along the axis are excluded by the symmetry $x_1 \mapsto -x_1$, and the Möbius automorphisms of the disc are excluded by requiring $\psi(0) = 0$ with real Taylor coefficients.

4. DISK POLYNOMIALS AND COEFFICIENT SPACES

4.1. Disk polynomials. Write $w = re^{i\theta}$ and $t = r^2$. For $m \in \mathbb{Z}$ and an integer $s \geq 0$ let

$$(4.1) \quad \Phi_{m,s}(w) = r^{|m|} P_s^{(0,|m|)}(2r^2 - 1) e^{im\theta},$$

where $P_s^{(\alpha,\beta)}$ is the Jacobi polynomial in the standard normalisation, so that $P_s^{(0,n)}(1) = 1$ and $\Phi_{m,s} = e^{im\theta}$ on $\partial\mathbb{D}$. Explicitly,

$$(4.2) \quad P_s^{(0,n)}(2t - 1) = \sum_{k=0}^s a_{s,k}^{(n)} t^k, \quad a_{s,k}^{(n)} = (-1)^{s-k} \frac{(n+s+k)!}{k!(s-k)!(n+k)!},$$

and, equivalently, $P_s^{(0,n)}(2t - 1) = \frac{1}{s!} t^{-n} \frac{d^s}{dt^s} \{t^{n+s}(t-1)^s\}$. We set $a_{s,k}^{(n)} = 0$ for $k \notin \{0, \dots, s\}$ and $\Phi_{m,-1} = 0$. Each $\Phi_{m,s}$ is a polynomial in $w, \bar{w}$ of degree $|m| + 2s$; these are the disk (Zernike) polynomials [28]. Complex conjugation gives $\overline{\Phi_{m,s}} = \Phi_{-m,s}$.

Lemma 4.1. *The disk polynomials have the following properties.*

- (a) For $n, s \geq 0$, the polynomial $P_s^{(0,n)}(2t - 1)$ is orthogonal in $L^2([0, 1], t^n dt)$ to all polynomials of degree less than s .
- (b) The functions $\Phi_{m,s}$ are pairwise orthogonal in $L^2(\mathbb{D})$, and $\int_{\mathbb{D}} |\Phi_{m,s}|^2 dA = \pi/(|m| + 2s + 1)$.
- (c) $|\Phi_{m,s}(w)| \leq 1$ for $|w| \leq 1$.

Proof. (a) Integrate the Rodrigues formula by parts s times against $t^n \cdot t^i$, $i < s$; all boundary terms vanish because the derivatives of order less than s of $t^{n+s}(t-1)^s$ vanish at $t = 0$ and $t = 1$.

(b) Angular integration separates different m . For equal m with $n = |m|$, $\int_{\mathbb{D}} \Phi_{m,s} \overline{\Phi_{m,s'}} dA = \pi \int_0^1 t^n P_s^{(0,n)} P_{s'}^{(0,n)} dt$, which vanishes for $s \neq s'$ by (a). For $s = s'$, by (a) only the leading

coefficient $a_{s,s}^{(n)} = \frac{(n+2s)!}{s!(n+s)!}$ of one factor contributes, and s integrations by parts in the Rodrigues formula give $\int_0^1 t^{n+s} P_s^{(0,n)} dt = \int_0^1 t^{n+s} (1-t)^s dt = \frac{(n+s)! s!}{(n+2s+1)!}$. The product is $1/(n+2s+1)$.

(c) Let $|w| \leq 1$, $h = \sqrt{1-|w|^2}$, and consider the unitary matrix $U = \begin{pmatrix} w & -h \\ h & \bar{w} \end{pmatrix}$, acting on the space of homogeneous polynomials of degree $N = n+2s$ in two variables e_1, e_2 , with the $U(2)$ -invariant inner product in which the monomials are orthogonal. The induced operator is unitary, so its diagonal matrix coefficients with respect to the normalised monomials have modulus at most one. The coefficient of $e_1^{n+s} e_2^s$ in $(we_1 + he_2)^{n+s} (-he_1 + \bar{w}e_2)^s$ is

$$\sum_{k=0}^s \binom{n+s}{k} \binom{s}{k} w^{n+s-k} \bar{w}^{s-k} (-h^2)^k = e^{in\theta} r^n \sum_{k=0}^s \binom{n+s}{k} \binom{s}{k} t^{s-k} (t-1)^k.$$

By Leibniz's rule applied to the Rodrigues formula, the last sum equals $P_s^{(0,n)}(2t-1)$. This proves (c) for $m = n \geq 0$; conjugation gives $m < 0$. $\square$

Fix $\rho > 1$ (we shall take $\rho = 21/20$). Let $\mathcal{B}_\rho$ be the space of complex sequences $f = (f_{m,s})_{m \in \mathbb{Z}, s \geq 0}$ with

$$\|f\|_\rho = \sum_{m,s} \rho^{|m|} |f_{m,s}| < \infty.$$

By Lemma 4.1(c), $\sum f_{m,s} \Phi_{m,s}$ converges absolutely and uniformly on $\overline{\mathbb{D}}$, with $\sup_{\overline{\mathbb{D}}} |f| \leq \|f\|_\rho$; by Lemma 4.1(b) the coefficients are determined by the function. We identify f with this continuous function. $\mathcal{B}_\rho$ is a Banach space, and every bounded linear operator on it (or between such weighted ℓ^1 spaces) is determined by its values on the basis vectors.

Lemma 4.2 (Multiplication by w and $\bar{w}$). *Let $n = |m|$. Then*

$$(4.3) \quad w \Phi_{m,s} = \begin{cases} \frac{n+s+1}{n+2s+1} \Phi_{m+1,s} + \frac{s}{n+2s+1} \Phi_{m+1,s-1}, & m \geq 0, \\ \frac{n+s}{n+2s+1} \Phi_{m+1,s} + \frac{s+1}{n+2s+1} \Phi_{m+1,s+1}, & m < 0, \end{cases}$$

and $\bar{w} \Phi_{m,s} = \overline{w \Phi_{-m,s}}$. In both cases the coefficients are nonnegative and sum to one. Consequently $\|wf\|_\rho \leq \rho \|f\|_\rho$ and $\|\bar{w}f\|_\rho \leq \rho \|f\|_\rho$, and for every holomorphic $p(w) = \sum_{a \geq 0} p_a w^a$ with $N_\rho(p) := \sum_a \rho^a |p_a| < \infty$,

$$(4.4) \quad \|pf\|_\rho \leq N_\rho(p) \|f\|_\rho, \quad \|\bar{p}f\|_\rho \leq N_\rho(p) \|f\|_\rho.$$

Moreover, if $m \geq 0$ then $w^N \Phi_{m,s}$ is a convex combination of $\Phi_{m+N,s'}$ with $s' \leq s$.

Proof. For $m = n \geq 0$ both sides of the first formula are $r^{n+1} e^{i(n+1)\theta}$ times a polynomial in t , and the claim is the identity $a_{s,k}^{(n)} = \frac{n+s+1}{n+2s+1} a_{s,k}^{(n+1)} + \frac{s}{n+2s+1} a_{s-1,k}^{(n+1)}$. Dividing by $a_{s,k}^{(n+1)}$ (which is nonzero for $0 \leq k \leq s$) and using (4.2), it reduces to $(n+s+1)(n+s+k+1) - s(s-k) = (n+k+1)(n+2s+1)$. For $m = -n < 0$ the claim is $a_{s,k-1}^{(n)} = \frac{n+s}{n+2s+1} a_{s,k}^{(n-1)} + \frac{s+1}{n+2s+1} a_{s+1,k}^{(n-1)}$; dividing by $a_{s,k}^{(n-1)}$ for $0 \leq k \leq s$ it reduces to $(n+s)(s+1-k) - (s+1)(n+s+k) = -k(n+2s+1)$, and the case $k = s+1$ is checked directly. The formula for $\bar{w}$ follows from $\overline{\Phi_{m,s}} = \Phi_{-m,s}$. Since $|m \pm 1| \leq |m| + 1$, the weight grows by at most a factor ρ , which gives the bounds for w and $\bar{w}$; (4.4) follows by iteration and absolute summation. The last assertion follows from the case $m \geq 0$ of (4.3). $\square$

Lemma 4.3 (Radial index under monomials). *Let $\alpha, \beta \geq 0$, $m \in \mathbb{Z}$, $s \geq 0$. Then $w^\alpha \bar{w}^\beta \Phi_{m,s}$ is a finite linear combination of $\Phi_{m+\alpha-\beta,s'}$ with $s' \geq s - \max(\alpha, \beta)$.*

Proof. Put $m' = m + \alpha - \beta$. The product equals $r^{\alpha+\beta+|m|} P_s^{(0,|m|)}(2t-1) e^{im'\theta}$, so, by Lemma 4.1(b) its coefficient on $\Phi_{m',s'}$ is a multiple of

$$\int_0^1 t^{|m|} P_s^{(0,|m|)}(2t-1) t^e P_{s'}^{(0,|m'|)}(2t-1) dt, \quad e = \frac{\alpha + \beta + |m'| - |m|}{2}.$$

Since $|m'| - |m| \equiv \alpha + \beta \pmod{2}$ and $||m'| - |m|| \leq |\alpha - \beta|$, e is an integer with $0 \leq \min(\alpha, \beta) \leq e \leq \max(\alpha, \beta)$. The second factor is a polynomial of degree $e + s'$, so the integral vanishes if $e + s' < s$ by Lemma 4.1(a). $\square$

Lemma 4.4 (Monomials). *Let $\alpha \geq \beta \geq 0$. Then $w^\alpha \bar{w}^\beta = \sum_{s=0}^\beta \kappa_s \Phi_{\alpha-\beta,s}$ with $\kappa_s \geq 0$, $\sum_s \kappa_s = 1$ and $\kappa_0 = \frac{\alpha - \beta + 1}{\alpha + 1}$.*

Proof. $\bar{w}^\beta = \Phi_{-\beta,0}$. Applying (4.3) α times preserves nonnegativity and total mass one, raises the radial index by at most one in each of the first β steps and never afterwards. The harmonic coefficient is the L^2 projection onto $r^{\alpha-\beta} e^{i(\alpha-\beta)\theta}$, namely $\int_0^1 r^{2\alpha+1} dr / \int_0^1 r^{2(\alpha-\beta)+1} dr$. $\square$

4.2. The sine sector. For odd $n \geq 1$ and $s \geq 0$ let

$$S_{n,s} = \operatorname{Im} \Phi_{n,s} = r^n P_s^{(0,n)}(2r^2 - 1) \sin n\theta = \frac{\Phi_{n,s} - \Phi_{-n,s}}{2i}.$$

Let $\mathcal{S}$ be the real Banach space of $f = \sum_{n \text{ odd}, s \geq 0} f_{n,s} S_{n,s}$ with $\|f\|_\rho = \sum \rho^n |f_{n,s}| < \infty$; this is also its norm as an element of $\mathcal{B}_\rho$. The elements of $\mathcal{S}$ are real-valued, odd under $w \mapsto \bar{w}$ and even under $w \mapsto -\bar{w}$. Conversely, since $\Phi_{m,s}(\bar{w}) = \Phi_{-m,s}(w)$ and $\Phi_{m,s}(-\bar{w}) = (-1)^m \Phi_{-m,s}(w)$, every real-valued $f \in \mathcal{B}_\rho$ with these two symmetries belongs to $\mathcal{S}$. The functions $S_{n,0} = r^n \sin n\theta$ are harmonic; we call $f_{n,0}$ the harmonic coefficients and put

$$\mathcal{G} = \{g \in \mathcal{S} : g_{n,0} = 0 \text{ for all } n\}.$$

5. THE COMPATIBLE INVERSE OF THE LAPLACIAN

For odd $n \geq 1$ and $s \geq 1$ put $q = n + 2s$ and define

$$(5.1) \quad KS_{n,s} = \frac{S_{n,s-1}}{4q(q+1)} - \frac{S_{n,s}}{2q(q+2)} + \frac{S_{n,s+1}}{4(q+1)(q+2)}.$$

Lemma 5.1. *For odd $n \geq 1$ and $s \geq 1$:*

- (a) $\Delta KS_{n,s} = S_{n,s}$;
- (b) $KS_{n,s} = 0$ and $\nabla KS_{n,s} = 0$ on $\partial\mathbb{D}$;
- (c) $KS_{n,s} = \frac{(1-r^2)^2}{4s(s+1)} r^n P_{s-1}^{(2,n)}(2r^2 - 1) \sin n\theta$;
- (d) *the absolute sum of the coefficients in (5.1) is $1/(q(q+2))$. Hence K extends to a bounded operator $K: \mathcal{G} \rightarrow \mathcal{S}$ with $\|K\| = 1/15$.*

Proof. Write A, B, C for the three coefficients in (5.1) and $a_{s,k} = a_{s,k}^{(n)}$.

(a) Since $\Delta(r^{n+2k} \sin n\theta) = 4k(n+k)r^{n+2k-2} \sin n\theta$, the claim is equivalent to $Aa_{s-1,k+1} + Ba_{s,k+1} + Ca_{s+1,k+1} = a_{s,k}/(4(k+1)(n+k+1))$ for $0 \leq k \leq s$. By (4.2) the three ratios $a_{s-1,k+1}/a_{s,k}$, $a_{s,k+1}/a_{s,k}$, $a_{s+1,k+1}/a_{s,k}$ equal $(s-k)(s-k-1)$, $-(n+s+k+1)(s-k)$ and $(n+s+k+2)(n+s+k+1)$, each divided by $(k+1)(n+k+1)$, and a direct computation gives $A(s-k)(s-k-1) - B(n+s+k+1)(s-k) + C(n+s+k+2)(n+s+k+1) = \frac{1}{4}$.

(b) At $r = 1$ each $r^n P_t^{(0,n)}(2r^2 - 1)$ equals one, and $A + B + C = 0$. Its radial derivative at $r = 1$ is $\beta_t = n + 2t(t+n+1)$, because $\frac{d}{dx} P_t^{(0,n)}(1) = t(t+n+1)/2$. Since $\beta_s - \beta_{s-1} = 2q$ and

$\beta_{s+1} - \beta_s = 2(q+2)$, we get $A\beta_{s-1} + B\beta_s + C\beta_{s+1} = -2qA + 2(q+2)C = 0$. The tangential derivative vanishes because the boundary value vanishes identically.

(c) By (b), $KS_{n,s} = r^n R(t) \sin n\theta$ where R is a polynomial of degree $s+1$ with $R(1) = 0$ and $nR(1) + 2R'(1) = 0$, so $R(t) = (1-t)^2 Q(t)$ with $\deg Q = s-1$. As a combination of $P_{s-1}^{(0,n)}, P_s^{(0,n)}, P_{s+1}^{(0,n)}$, the polynomial R is orthogonal in $L^2(t^n dt)$ to polynomials of degree less than $s-1$ (Lemma 4.1(a)); hence Q is orthogonal in $L^2(t^n(1-t)^2 dt)$ to them and is a multiple of $P_{s-1}^{(2,n)}(2t-1)$. Comparing leading coefficients, $C a_{s+1,s+1}$ for R and $\frac{(n+2s)!}{(s-1)!(n+s+1)!}$ for $P_{s-1}^{(2,n)}(2t-1)$, gives the factor $1/(4s(s+1))$.

(d) $A - B + C = \frac{(q+2)+2(q+1)+q}{4q(q+1)(q+2)} = \frac{1}{q(q+2)}$. The operator K does not change the angular index, hence not the weight. Since $q \geq 3$, the supremum of $1/(q(q+2))$ is $1/15$, attained at $(n, s) = (1, 1)$. $\square$

Lemma 5.2. *For $g \in \mathcal{G}$, the function $W = Kg$ belongs to $C^1(\overline{\mathbb{D}})$, satisfies $\Delta W = g$ in $\mathcal{D}'(\mathbb{D})$, and $W = 0, \nabla W = 0$ on $\partial\mathbb{D}$.*

Proof. Let g_N be the truncations of g to finitely many coefficients and $W_N = Kg_N$, a polynomial with $\Delta W_N = g_N$ and zero Cauchy data by Lemma 5.1. The Dirichlet Green function of the disc is $G(w, \zeta) = \frac{1}{2\pi} \log \left| \frac{w-\zeta}{1-w\bar{\zeta}} \right|$, and $W_N(w) = \int_{\mathbb{D}} G(w, \zeta) g_N(\zeta) dA(\zeta)$. We have $|\nabla_w G(w, \zeta)| \leq \frac{1}{2\pi} (|w-\zeta|^{-1} + |\zeta| |1-w\bar{\zeta}|^{-1})$. For $|w| \leq 1$, $\int_{\mathbb{D}} |w-\zeta|^{-1} dA(\zeta) \leq 4\pi$, since $\mathbb{D}$ lies in the disc of radius 2 about w . If $|w| < 1/2$ then $|\zeta|/|1-w\bar{\zeta}| \leq 2$; if $|w| \geq 1/2$ then $|\zeta|/|1-w\bar{\zeta}| \leq 2/|\zeta - 1/\bar{w}|$ and $\mathbb{D}$ lies in the disc of radius 3 about $1/\bar{w}$; in both cases the integral of this term is at most 12π . Hence

$$\sup_{\mathbb{D}} |\nabla(W_N - W_L)| \leq 8 \sup_{\mathbb{D}} |g_N - g_L| \leq 8 \|g_N - g_L\|_{\rho}, \quad \sup_{\mathbb{D}} |W_N - W_L| \leq \frac{1}{15} \|g_N - g_L\|_{\rho}.$$

Thus (W_N) is Cauchy in $C^1(\overline{\mathbb{D}})$; its limit is Kg (the coefficients converge in $\mathcal{S}$), the zero Cauchy data pass to the limit, and so does the distributional equation. $\square$

Remark 5.3. Conversely, if $W \in C^1(\overline{\mathbb{D}})$ has $\Delta W = g \in \mathcal{S}$ in $\mathcal{D}'(\mathbb{D})$ and zero Cauchy data, then $g \in \mathcal{G}$ and $W = Kg$: the extension of W by zero is C^1 and satisfies $\Delta W = g \mathbf{1}_{\mathbb{D}}$ in $\mathcal{D}'(\mathbb{R}^2)$, and pairing with the harmonic polynomials $r^n \sin n\theta$ gives $g_{n,0} = 0$; then $W - Kg$ is harmonic with zero boundary values. This is why the equation is posed on $\mathcal{G}$.

6. THE CUBIC EQUATION

From now on $\rho = 21/20$, $M = 41$ and $S = 24$. The centre $x^\circ = (g^\circ, c^\circ)$ consists of 504 coefficients $g_{n,s}^\circ$, $n \in \{1, 3, \dots, 41\}$, $1 \leq s \leq 24$, and 21 coefficients c_j° , $j \in \{1, 3, \dots, 41\}$; all other coefficients of x° vanish. Each of these 525 numbers is an explicitly given dyadic rational (a binary64 number interpreted exactly). We write $b = c_1^\circ \approx 14.840043$ and

$$\omega_j = b^2(j+1)\rho^j \quad (j \text{ odd}).$$

Let X be the real Banach space of pairs $x = (g, c)$, with $g \in \mathcal{G}$ and $c = (c_j)_{j \text{ odd} \geq 1}$ real, normed by

$$\|(g, c)\|_X = \|g\|_{\rho} + \sum_{j \text{ odd}} \omega_j |c_j|.$$

To c we associate $\psi_c(w) = \sum_j c_j w^j$, which is holomorphic in $|w| < \rho$ and continuous up to $|w| = \rho$ because $|c_j| \rho^j \leq \omega_j |c_j| / (2b^2)$. Note that $\text{Im } \psi_c = \sum_j c_j S_{j,0} \in \mathcal{S}$. Define

$$(6.1) \quad \mathcal{F}(g, c) = g + |\psi'_c|^2 (Kg + \text{Im } \psi_c).$$

Lemma 6.1. $\mathcal{F}$ is a continuous cubic polynomial map $X \rightarrow \mathcal{S}$. If $\mathcal{F}(g, c) = 0$, then $V = Kg + \text{Im } \psi_c$ belongs to $C^1(\mathbb{D})$, is odd under $w \mapsto \bar{w}$ and even under $w \mapsto -\bar{w}$, and satisfies (3.1) with $\psi = \psi_c$.

Proof. $\psi'_c = \sum_j j c_j w^{j-1}$ has real coefficients and even powers, and $N_\rho(\psi'_c) = \sum_j j \rho^{j-1} |c_j| \leq \|c\|/(b^2 \rho)$, where $\|c\| = \sum \omega_j |c_j|$. Thus $|\psi'_c|^2$ is real and invariant under $w \mapsto \bar{w}$ and $w \mapsto -\bar{w}$, and multiplication by it maps $\mathcal{S}$ into $\mathcal{S}$ with norm at most $N_\rho(\psi'_c)^2$ by (4.4). Together with $\|Kg\|_\rho \leq \|g\|_\rho/15$ and $\|\text{Im } \psi_c\|_\rho = \sum \rho^j |c_j|$, this shows that (6.1) is a sum of bounded multilinear maps of degree at most three. If $\mathcal{F}(g, c) = 0$, then Lemma 5.2 gives $V \in C^1(\mathbb{D})$, $V - \text{Im } \psi_c = Kg$ with zero Cauchy data, and $\Delta V = g = -|\psi'_c|^2 V$ in $\mathcal{D}'(\mathbb{D})$; the last identity holds pointwise because both sides are uniformly convergent disk-polynomial series with the same coefficients. The parities hold because $V \in \mathcal{S}$. $\square$

6.1. Expansion at the centre. Write $p = \psi'_{c^\circ} = \sum_{a=0,2,\dots,40} p_a w^a$ with $p_a = (a+1)c_{a+1}^\circ$, so $p_0 = b$, and put

$$P = N_\rho(p), \quad V^\circ = Kg^\circ + \text{Im } \psi_{c^\circ}, \quad V_0 = \|V^\circ\|_\rho.$$

For $h = (\delta g, \eta) \in X$ let $\delta V = K\delta g + \text{Im } \psi_\eta$ and, for shape perturbations η, η_1, η_2 ,

$$L(\eta) = \bar{p} \psi'_\eta + p \overline{\psi'_\eta} = 2 \text{Re}(\bar{p} \psi'_\eta), \quad Q(\eta_1, \eta_2) = \text{Re}(\psi'_{\eta_1} \overline{\psi'_{\eta_2}}).$$

Since $|p + \psi'_\eta|^2 = |p|^2 + L(\eta) + Q(\eta, \eta)$, expanding (6.1) gives

$$(6.2) \quad \mathcal{F}(x^\circ + h) = \mathcal{F}(x^\circ) + D\mathcal{F}(x^\circ)h + B_2(h, h) + B_3(h, h, h),$$

where

$$(6.3) \quad \begin{aligned} D\mathcal{F}(x^\circ)h &= \delta g + |p|^2 K\delta g + L(\eta)V^\circ + |p|^2 \text{Im } \psi_\eta, \\ B_2(h_1, h_2) &= \frac{1}{2} \{ L(\eta_1) \delta V_2 + L(\eta_2) \delta V_1 \} + Q(\eta_1, \eta_2)V^\circ, \\ B_3(h_1, h_2, h_3) &= \frac{1}{3} \{ Q(\eta_1, \eta_2) \delta V_3 + Q(\eta_1, \eta_3) \delta V_2 + Q(\eta_2, \eta_3) \delta V_1 \}. \end{aligned}$$

There are no terms of higher order.

Lemma 6.2. Let $\kappa = 1/15$, $\vartheta_1 = 1/(b^2 \rho)$ and $\vartheta_0 = 1/(2b^2)$. Then for all $h_i \in X$,

$$\|B_2(h_1, h_2)\|_\rho \leq c_2 \|h_1\|_X \|h_2\|_X, \quad \|B_3(h_1, h_2, h_3)\|_\rho \leq c_3 \|h_1\|_X \|h_2\|_X \|h_3\|_X,$$

with $c_2 = \max\{P\vartheta_1\kappa, 2P\vartheta_1\vartheta_0 + \vartheta_1^2 V_0\}$ and $c_3 = \vartheta_1^2 \max\{\kappa, \vartheta_0\}$.

Proof. Write $a_i = \|\delta g_i\|_\rho$ and $e_i = \sum_j \omega_j |\eta_{i,j}|$. Termwise, $N_\rho(\psi'_{\eta_i}) \leq \vartheta_1 e_i$ (as $j/(j+1) \leq 1$), $\|\text{Im } \psi_{\eta_i}\|_\rho \leq \vartheta_0 e_i$ (as $j+1 \geq 2$), and $\|K\delta g_i\|_\rho \leq \kappa a_i$, so $\|\delta V_i\|_\rho \leq \kappa a_i + \vartheta_0 e_i$. By (4.4), $\|L(\eta_i)f\|_\rho \leq 2P\vartheta_1 e_i \|f\|_\rho$ and $\|Q(\eta_i, \eta_k)f\|_\rho \leq \vartheta_1^2 e_i e_k \|f\|_\rho$. Hence $\|B_2(h_1, h_2)\|_\rho \leq P\vartheta_1\kappa(e_1 a_2 + e_2 a_1) + (2P\vartheta_1\vartheta_0 + \vartheta_1^2 V_0)e_1 e_2 \leq c_2(a_1 + e_1)(a_2 + e_2)$. Similarly every monomial $e_i e_k a_l$ in the bound for B_3 has coefficient $\vartheta_1^2 \kappa/3$ and $e_1 e_2 e_3$ has coefficient $\vartheta_1^2 \vartheta_0$; each is at most its coefficient in $c_3 \prod_i (a_i + e_i)$. $\square$

7. THE CONTRACTION ARGUMENT AND THE APPROXIMATE INVERSE

7.1. A Newton–Kantorovich theorem.

Proposition 7.1. Let $\mathcal{F}: X \rightarrow \mathcal{S}$ have the form (6.2) with symmetric bounded B_2 and B_3 , and let $A: \mathcal{S} \rightarrow X$ be bounded, linear and injective. Suppose that

$$\|A\mathcal{F}(x^\circ)\|_X \leq Y, \quad \|I - A D\mathcal{F}(x^\circ)\|_{X \rightarrow X} \leq Z, \quad \|A\|_{c_2} \leq C_2, \quad \|A\|_{c_3} \leq C_3,$$

and that $r > 0$ satisfies

$$(7.1) \quad Y + (Z - 1)r + C_2 r^2 + C_3 r^3 < 0, \quad Z - 1 + 2C_2 r + 3C_3 r^2 < 0.$$

Then $\mathcal{F}$ has exactly one zero in the closed ball $\overline{B}_X(x^\circ, r)$.

Proof. Let $\mathcal{N}(x) = x - A\mathcal{F}(x)$. For $\|h\|_X \leq r$, (6.2) gives $\mathcal{N}(x^\circ + h) - x^\circ = -A\mathcal{F}(x^\circ) + (I - A\mathcal{D}\mathcal{F}(x^\circ))h - AB_2(h, h) - AB_3(h, h, h)$, whose norm is at most $Y + Zr + C_2r^2 + C_3r^3 < r$. For h_1, h_2 in the ball, $B_2(h_1, h_1) - B_2(h_2, h_2) = B_2(h_1 - h_2, h_1 + h_2)$ and $B_3(h_1, h_1, h_1) - B_3(h_2, h_2, h_2) = B_3(h_1 - h_2, h_1, h_1) + B_3(h_2, h_1 - h_2, h_1) + B_3(h_2, h_2, h_1 - h_2)$, so $\mathcal{N}$ is Lipschitz on the ball with constant $Z + 2C_2r + 3C_3r^2 < 1$. By the contraction mapping principle $\mathcal{N}$ has a unique fixed point in the ball, and since A is injective the fixed points of $\mathcal{N}$ are exactly the zeros of $\mathcal{F}$. $\square$

In the weighted ℓ^1 spaces X and $\mathcal{S}$, operator norms are computed column by column.

Lemma 7.2. *Let E and E' be weighted ℓ^1 spaces with bases (e_i) and (e'_κ) and weights (λ_i) and (λ'_κ) , and let $T: E \rightarrow E'$ be bounded. Then $\|T\| = \sup_i \|Te_i\|_{E'}/\lambda_i$.*

Proof. The inequality $\geq$ follows by testing on e_i/λ_i , and $\leq$ by the triangle inequality and absolute summation, using continuity of T and density of finitely supported vectors. $\square$

We shall apply Lemma 7.2 with the bases $\{e_{(n,s)} : s \geq 1\} \cup \{e_j\}$ of X , of weights ρ^n and ω_j , and $\{S_{n,s} : s \geq 0\}$ of $\mathcal{S}$, of weights ρ^n . We call the basis vectors of X *inputs* and those of $\mathcal{S}$ *rows*.

7.2. Finite block and tail. The *finite inputs* are the 504 coordinates $g_{n,s}$ with $n \leq M$, $1 \leq s \leq S$, and the 21 shape coordinates c_j with $j \leq M$. The *finite rows* are the 504 rows (n, s) with $n \leq M$, $1 \leq s \leq S$, and the 21 harmonic rows $(n, 0)$ with $n \leq M$. All other inputs and rows are *tail* inputs and rows. We write $X = X_f \oplus X_t$ and $\mathcal{S} = \mathcal{S}_f \oplus \mathcal{S}_t$ accordingly; $\dim X_f = \dim \mathcal{S}_f = 525$. Let M_f be the 525×525 matrix of the restriction of $D\mathcal{F}(x^\circ)$ to finite inputs and finite rows.

On tail inputs we use the following explicit model of $D\mathcal{F}(x^\circ)$, obtained by replacing ψ° by bw and V° by $b \operatorname{Im} w$:

$$(7.2) \quad T(\delta g, \eta) = \delta g + b^2 \sum_{j > M} \eta_j \{ (j+1)S_{j,0} - (j-1)S_{j-2,0} - S_{j-2,1} \}.$$

Lemma 7.3. *For odd $j \geq 3$, $L_b(w^j) b \operatorname{Im} w + b^2 \operatorname{Im} w^j = b^2 \{ (j+1)S_{j,0} - (j-1)S_{j-2,0} - S_{j-2,1} \}$, where $L_b(\eta) = 2 \operatorname{Re}(b\psi'_\eta)$.*

Proof. $L_b(w^j) b \operatorname{Im} w = 2b^2 j r^j \cos((j-1)\theta) \sin \theta = b^2 j r^j \{ \sin j\theta - \sin(j-2)\theta \}$. Moreover $r^2 \Phi_{j-2,0} = \frac{1}{j} \Phi_{j-2,1} + \frac{j-1}{j} \Phi_{j-2,0}$, because $P_1^{(0,n)}(2t-1) = (n+2)t - (n+1)$ by (4.2); hence $j r^j \sin(j-2)\theta = S_{j-2,1} + (j-1)S_{j-2,0}$. $\square$

Write $Te = (Be, T_{tt}e)$ for tail inputs e , with B the part in finite rows and T_{tt} the part in tail rows. Only the shape input $j = M+2$ reaches finite rows:

$$(7.3) \quad B(\delta g, \eta) = -b^2 \eta_{M+2} ((M+1)e_{(M,0)} + e_{(M,1)}).$$

Put $\tau = (1 - \rho^{-2})^{-1}$.

Lemma 7.4. *$T_{tt}: X_t \rightarrow \mathcal{S}_t$ is a bounded bijection. For a tail residual $y = \sum y_{n,s} S_{n,s} \in \mathcal{S}_t$ its inverse is*

$$(7.4) \quad \eta_n = \sum_{\substack{m \geq n \\ m \text{ odd}}} \frac{y_{m,0}}{b^2(m+1)} \quad (n > M), \quad \delta g_{n,s} = y_{n,s} + \mathbf{1}_{s=1} b^2 \eta_{n+2} \quad ((n, s) \text{ a tail } g\text{-input}).$$

Moreover $\|T_{tt}\| \leq 1 + \rho^{-2}$ and $\|T_{tt}^{-1}\| \leq \max\{1, \tau + \tau/((M+3)\rho^2)\}$.

Proof. By (7.2), the tail harmonic rows read $b^2(n+1)(\eta_n - \eta_{n+2}) = y_{n,0}$ for $n \geq M+2$, the tail rows $(n, 1)$ read $\delta g_{n,1} - b^2\eta_{n+2} = y_{n,1}$, and all other tail rows are $\delta g_{n,s} = y_{n,s}$. Since $\sum \omega_n |\eta_n| < \infty$ forces $\eta_n \rightarrow 0$, summation of the harmonic equations gives (7.4), and a solution with $y = 0$ vanishes; conversely (7.4) solves the equations whenever the series converge. For the bounds we use Lemma 7.2. The shape input e_j has three nonzero entries, $b^2(j+1)$ in row $(j, 0)$, $-b^2(j-1)$ in row $(j-2, 0)$ and $-b^2$ in row $(j-2, 1)$; its weighted norm divided by ω_j is $1 + j\rho^{-2}/(j+1) \leq 1 + \rho^{-2}$. Tail g -inputs are mapped to the identical rows. For T_{tt}^{-1} , a nonharmonic tail row is mapped to the identical input. The normalised harmonic row $\rho^{-m}S_{m,0}$ is mapped to $\eta_n = \rho^{-m}/(b^2(m+1))$ for $M < n \leq m$ and $\delta g_{n-2,1} = b^2\eta_n$ for $M+4 \leq n \leq m$, with norm

$$\sum_{M < n \leq m} \frac{(n+1)\rho^n}{(m+1)\rho^m} + \sum_{M+4 \leq n \leq m} \frac{\rho^{n-2}}{(m+1)\rho^m} \leq \tau + \frac{\tau}{(M+3)\rho^2}. \quad \square$$

7.3. The approximate inverse. Let $\hat{A}$ be a 525×525 matrix with dyadic rational entries (in the computation, the midpoints of an interval enclosure of M_f^{-1}), regarded as an exact linear map $\mathcal{S}_f \rightarrow X_f$, and suppose that

$$(7.5) \quad \|I - \hat{A}M_f\|_{X_f \rightarrow X_f} < 1.$$

Define $A: \mathcal{S} = \mathcal{S}_f \oplus \mathcal{S}_t \rightarrow X = X_f \oplus X_t$ by

$$(7.6) \quad A = \begin{pmatrix} \hat{A} & -\hat{A}BT_{tt}^{-1} \\ 0 & T_{tt}^{-1} \end{pmatrix}.$$

Lemma 7.5. *Under (7.5), A is bounded and injective, and $ATe = e$ for every tail input e .*

Proof. By (7.5) and the Neumann series, $\hat{A}M_f$ is invertible, so $\hat{A}$ is invertible. The operator $\begin{pmatrix} \hat{A}^{-1} & B \\ 0 & T_{tt} \end{pmatrix}$ is a left inverse of A . For a tail input e , $ATe = (\hat{A}Be - \hat{A}BT_{tt}^{-1}T_{tt}e, T_{tt}^{-1}T_{tt}e) = (0, e)$. $\square$

Let $e_{(n,s)}$ denote the finite rows as elements of $\mathcal{S}_f$, and define

$$\|R\| = \max_{(n,s) \text{ finite row}} \frac{\|\hat{A}e_{(n,s)}\|_{X_f}}{\rho^n}, \quad H = \|\hat{A}(e_{(M,1)} + (M+1)e_{(M,0)})\|_{X_f},$$

and, for odd $n \geq M+2$,

$$(7.7) \quad h_n = \frac{H + \sum_{j=M+2, M+4, \dots, n} (j+1)\rho^j + \sum_{j=M+4, M+6, \dots, n} \rho^{j-2}}{(n+1)\rho^n}.$$

Lemma 7.6. *Let $N_H \geq M+2$ be odd. Then A maps every residual supported in tail rows with norm at most*

$$A_t := \max \left\{ 1, \max_{M+2 \leq n \leq N_H} h_n, \tau + \frac{\tau}{(N_H+3)\rho^2} + \frac{H}{(N_H+3)\rho^{N_H+2}} \right\},$$

it is the identity on residuals supported in nonharmonic tail rows, and $\|A\| \leq \max\{\|R\|, A_t\}$.

Proof. We use Lemma 7.2. A finite row $e_{(n,s)}$ is mapped to $\hat{A}e_{(n,s)}$. A nonharmonic tail row is mapped to the identical tail input by (7.4), and B vanishes on it. The normalised harmonic tail row $\rho^{-n}S_{n,0}$ is mapped to the shape coordinates $\eta_j = \rho^{-n}/(b^2(n+1))$, $M+2 \leq j \leq n$, to $\delta g_{j-2,1} = b^2\eta_j$, $M+4 \leq j \leq n$, and, by (7.3), to the finite part $b^2\eta_{M+2}\hat{A}((M+1)e_{(M,0)} + e_{(M,1)})$; the sum of the three norms is exactly h_n . For $n \geq N_H+2$ we bound $(j+1)/(n+1) \leq 1$ in the first sum, sum the geometric series $\sum_{i \geq 0} \rho^{-2i} = \tau$, and use $n+1 \geq N_H+3$ in the other two terms. $\square$

8. THE DEFECT BOUNDS

In this section we explain how the bounds Y and Z of Proposition 7.1 are obtained for the operator A of (7.6). By Lemma 7.2, $\|I - AD\mathcal{F}(x^\circ)\|$ is the supremum over all inputs e of $\|(I - AD\mathcal{F}(x^\circ))e\|_X$ divided by the weight of e . We partition the inputs into five classes and bound each class separately; every input belongs to exactly one class, and no column is estimated by sampling. Throughout, $d = \deg p = 40$, and $\Lambda = M + d = 81$.

8.1. The residual and the finite columns. The residual $\mathcal{F}(x^\circ)$ is a polynomial in $w, \bar{w}$. Using (4.3) and (5.1), its expansion in the basis $S_{n,s}$ is computed without truncation (it has 2286 coefficients) in interval arithmetic. Only finitely many tail rows occur, so $A\mathcal{F}(x^\circ)$ is computed without truncation from (7.6), (7.3) and (7.4), and Y is an upper bound of its norm.

For each of the 525 finite inputs e , $D\mathcal{F}(x^\circ)e$ is again a polynomial by (6.3), and $(I - AD\mathcal{F}(x^\circ))e$ is computed in the same way. We denote by Z_f an upper bound for the maximum of the normalised norms of these columns.

8.2. Tail g -columns. Let $n_{\max} = M + 2d + 2 = 123$ and $s_{\max} = S + d + 3 = 67$.

Lemma 8.1. *Let $e = e_{(n,s)}$ be a tail g -input and $q = n + 2s$. Then $(I - AD\mathcal{F}(x^\circ))e = -A(|p|^2 K S_{n,s})$. Moreover, writing $p = p_{\leq 16} + p_{> 16}$ for the splitting of p into terms of degree ≤ 16 and > 16 , and $P_s = N_\rho(p_{\leq 16})$, $P_t = N_\rho(p_{> 16})$,*

$$(8.1) \quad \frac{\|(I - AD\mathcal{F}(x^\circ))e\|_X}{\rho^n} \leq \frac{\|A(|p_{\leq 16}|^2 K S_{n,s})\|_X}{\rho^n} + \frac{\|A\| (2P_s P_t + P_t^2)}{q(q+2)}.$$

Proof. By (6.3), $D\mathcal{F}(x^\circ)e = S_{n,s} + |p|^2 K S_{n,s}$, and $Te = S_{n,s}$, so $AS_{n,s} = ATe = e$ by Lemma 7.5. Since $|p|^2 - |p_{\leq 16}|^2 = \overline{p_{\leq 16}} p_{> 16} + \overline{p_{> 16}} p_{\leq 16} + |p_{> 16}|^2$, (4.4) and Lemma 5.1(d) give $\|(|p|^2 - |p_{\leq 16}|^2) K S_{n,s}\|_\rho \leq (2P_s P_t + P_t^2) \rho^n / (q(q+2))$. $\square$

For the $62 \cdot 67 - 21 \cdot 24 = 3650$ tail g -inputs with $n \leq n_{\max}$ and $s \leq s_{\max}$, the first term on the right of (8.1) is computed without truncation; let $Z_{g,\partial}$ bound the maximum of the right-hand side over these inputs.

Lemma 8.2. *For every tail g -input $e_{(n,s)}$ with $n > n_{\max}$ or $s > s_{\max}$,*

$$\frac{\|(I - AD\mathcal{F}(x^\circ))e_{(n,s)}\|_X}{\rho^n} \leq Z_{g,\infty} := \max \left\{ \frac{P^2}{137 \cdot 139}, \frac{A_t P^2}{127 \cdot 129} \right\}.$$

Proof. We have $|p|^2 = \sum_{a,a'} p_a p_{a'} w^a \bar{w}^{a'}$ with $a, a' \leq d$. By (5.1) and Lemma 4.3, every component of $|p|^2 K S_{n,s}$ has radial index at least $s - 1 - d$ and angular frequency of modulus at least $n - d$. Also $\||p|^2 K S_{n,s}\|_\rho \leq P^2 \rho^n / (q(q+2))$ with $q = n + 2s$.

If $s > s_{\max}$, all radial indices are at least $s - 1 - d \geq S + 3$, so all components lie in nonharmonic tail rows, on which A is the identity (Lemma 7.6); and $q \geq 1 + 2(s_{\max} + 1) = 137$.

If $s \leq s_{\max}$ and $n > n_{\max}$, all frequencies are at least $n - d \geq n_{\max} + 2 - d > M$, so all components lie in tail rows, on which A has norm at most A_t ; and $q \geq n_{\max} + 2 + 2 = 127$. $\square$

8.3. Shape columns. For the 480 shape inputs e_j with $M < j \leq j_* = 1001$, $(I - AD\mathcal{F}(x^\circ))e_j$ is computed without truncation as in Section 8.1; let $Z_{\text{sh},\partial}$ bound their normalised norms. For $j > j_*$ we prove a uniform bound.

Let $W^\circ = Kg^\circ$, so that $V^\circ = W^\circ + \text{Im } \psi_{c^\circ}$. By (6.3) and Lemma 7.3, for odd $j > M$,

$$D\mathcal{F}(x^\circ)e_j = 2 \text{Re} \left[w^{j-1} \left(j \bar{p} V^\circ + \frac{|p|^2 w}{2i} \right) \right], \quad Te_j = 2 \text{Re} \left[w^{j-1} \left(j b^2 \text{Im } w + \frac{b^2 w}{2i} \right) \right].$$

Since $ATe_j = e_j$, we get $(I - AD\mathcal{F}(x^\circ))e_j = -A\Delta_j$ with

$$(8.2) \quad \Delta_j = 2 \operatorname{Re} [w^{j-1}(j\Gamma_1 + \Gamma_0)] + 2 \operatorname{Re} [jw^{j-1}\bar{p}W^\circ],$$

where $\Gamma_1 = \bar{p} \operatorname{Im} \psi_{c^\circ} - b^2 \operatorname{Im} w$ and $\Gamma_0 = (|p|^2 - b^2)w/(2i)$. Writing $\operatorname{Im} \psi_{c^\circ} = (\psi_{c^\circ} - \overline{\psi_{c^\circ}})/(2i)$ and $|p|^2 = \sum p_a p_{a'} w^a \bar{w}^{a'}$, and using $p_0 = c_1^\circ = b$,

$$2i\Gamma_1 = \sum_{(a,l) \neq (0,1)} p_a c_l^\circ (w^l \bar{w}^a - \bar{w}^{a+l}), \quad 2i\Gamma_0 = \sum_{(a,a') \neq (0,0)} p_a p_{a'} w^{a+1} \bar{w}^{a'},$$

with $a, a' \in \{0, 2, \dots, 40\}$ and $l \in \{1, 3, \dots, 41\}$. After combining equal monomials, $2i\Gamma_1 = \sum_{\alpha, \beta} q_{\alpha\beta}^1 w^\alpha \bar{w}^\beta$ and $2i\Gamma_0 = \sum_{\alpha, \beta} q_{\alpha\beta}^0 w^\alpha \bar{w}^\beta$ with finitely many real coefficients. The differences $k = \alpha - \beta$ that occur are odd and lie in $[k_-, k_+] = [-81, 41]$. For odd $k \in [k_-, k_+]$ define

$$\begin{aligned} \bar{q}_k &= \sum_{\alpha-\beta=k} q_{\alpha\beta}^1, & \varepsilon_k &= \frac{1}{j_* + 1} \sum_{\alpha-\beta=k} ((1+\beta)|q_{\alpha\beta}^1| + |q_{\alpha\beta}^0|), & \bar{\eta}_k &= \frac{1}{b^2} \sum_{l \geq k} \bar{q}_l, \\ \delta_k &= \frac{1}{b^2} \sum_{l \geq k} \left[\varepsilon_l \left(1 + \frac{|l-1|}{j_* + k_-} \right) + \frac{|\bar{q}_l| |l-1|}{j_* + k_-} \right], & E_k &= |\bar{\eta}_k| + \delta_k, \end{aligned}$$

where the sums over l run over odd $l \leq k_+$. By Lemma 5.1(c), $W^\circ = (1 - |w|^2)^2 Q^\circ$ with

$$Q^\circ = \sum_{n,s} \frac{g_{n,s}^\circ}{4s(s+1)} r^n P_{s-1}^{(2,n)} (2r^2 - 1) \sin n\theta,$$

and Q° has a finite disk-polynomial expansion, which we compute from the connection formulas (they follow from [16, Eq. 18.9.5] and the symmetry $P_k^{(\alpha,\beta)}(-x) = (-1)^k P_k^{(\beta,\alpha)}(x)$, [16, Table 18.6.1])

$$P_k^{(a+1,n)} = \frac{2k+a+n+1}{k+a+n+1} P_k^{(a,n)} + \frac{k+n}{k+a+n+1} P_{k-1}^{(a+1,n)} \quad (a = 0, 1).$$

The function $w^\Lambda \bar{p}Q^\circ$ has only nonnegative frequencies, since the frequencies of $\bar{p}Q^\circ$ are at least $-M - d$; write $w^\Lambda \bar{p}Q^\circ = \sum_{m \geq 0, t \geq 0} \varpi_{m,t} \Phi_{m,t}$ (a finite sum, computed in interval arithmetic). Finally let

$$\begin{aligned} U_1 &= \sum_k \left(1 + \frac{\max(0, k-1)}{j_* + 1} \right) \rho^{k-1} E_k, & U_2 &= E_{k_-} \rho^{k_- - 3} \tau, \\ U_3 &= \frac{1}{j_* + 1} \left(\sum_k \rho^{k-3} E_k + E_{k_-} \rho^{k_- - 5} \tau \right), & U_4 &= \frac{H E_{k_-}}{(j_* + 1) \rho^{j_*}}, \\ U_5 &= \sum_{\alpha, \beta} \frac{(|q_{\alpha\beta}^1| + |q_{\alpha\beta}^0|/j_*) \beta \rho^{\alpha-\beta-1}}{b^2(j_* + \alpha)}, \\ U_W &= \frac{8}{b^2(j_* - \Lambda)^2} \sum_{m,t} |\varpi_{m,t}| (2t+1)(2t+3) \rho^{m-1-\Lambda}, \end{aligned}$$

where k runs over odd integers in $[k_-, k_+]$.

Lemma 8.3. *For integers $m, t, N \geq 0$,*

$$\|(1 - |w|^2)^2 w^N \Phi_{m,t}\|_\rho \leq \frac{4(2t+1)(2t+3)}{(m+N+1)^2} \rho^{m+N}.$$

Proof. Applying (4.3) for w and then for $\bar{w}$ gives $|w|^2\Phi_{m,t} = \lambda_t^+\Phi_{m,t+1} + (1 - \lambda_t^+ - \lambda_t^-)\Phi_{m,t} + \lambda_t^-\Phi_{m,t-1}$ with

$$\lambda_t^+ = \frac{(t+1)(m+t+1)}{(m+2t+1)(m+2t+2)} \leq \frac{t+1}{m+1}, \quad \lambda_t^- = \frac{t(m+t)}{(m+2t)(m+2t+1)} \leq \frac{t}{m+1}.$$

Hence $(1 - |w|^2)\Phi_{m,t}$ has the same frequency and absolute coefficient sum $2(\lambda_t^+ + \lambda_t^-) \leq 2(2t+1)/(m+1)$, and its radial indices are at most $t+1$. Applying this twice, $\|(1 - |w|^2)^2\Phi_{m,t}\|_\rho \leq 4(2t+1)(2t+3)\rho^m/(m+1)^2$. By Lemma 4.2, $w^N\Phi_{m,t}$ is a convex combination of $\Phi_{m+N,t'}$ with $t' \leq t$; apply the previous bound to each term. $\square$

Proposition 8.4. *Assume $j_* + k_- - 1 > M + 2$ and $j_* > 2M + d + 2$. Then for every odd $j > j_*$,*

$$\frac{\|(I - AD\mathcal{F}(x^\circ))e_j\|_X}{\omega_j} \leq Z_{\text{sh},\infty} := \mathbf{U}_1 + \mathbf{U}_2 + \mathbf{U}_3 + \mathbf{U}_4 + \mathbf{U}_5 + A_t \mathbf{U}_W.$$

Proof. Fix an odd $j > j_*$. For real γ , $2\operatorname{Re}[w^{j-1}\gamma w^\alpha \bar{w}^\beta/(2i)] = \gamma \operatorname{Im}(w^{j-1+\alpha}\bar{w}^\beta)$. For every monomial occurring, $k = \alpha - \beta \geq k_-$ and $j - 1 + k > M + 2 \geq 1$, so by Lemma 4.4 $\operatorname{Im}(w^{j-1+\alpha}\bar{w}^\beta) = \sum_s \kappa_s S_{j-1+k,s}$ with nonnegative coefficients summing to one and harmonic coefficient $\kappa_0 = 1 - \beta/(j + \alpha)$. Accordingly we split $\Delta_j = \Delta^H + \Delta^N + \Delta^W$ into the harmonic part and the nonharmonic part of the first term of (8.2), and the second term Δ^W . All rows occurring in Δ^H and Δ^N have frequency at least $j - 1 + k_- > M + 2$, so they are tail rows.

The harmonic part. The coefficient of Δ^H in row $(j - 1 + k, 0)$ is $y_k = \sum_{\alpha-\beta=k} (jq_{\alpha\beta}^1 + q_{\alpha\beta}^0)(1 - \frac{\beta}{j+\alpha})$. Since $\frac{j}{j+1}(1 - \frac{\beta}{j+\alpha}) - 1 = -\frac{1}{j+1} - \frac{j\beta}{(j+1)(j+\alpha)}$ and $0 \leq 1 - \frac{\beta}{j+\alpha} \leq 1$, we have $|y_k/(j+1) - \bar{q}_k| \leq \varepsilon_k$. By (7.4), the shape part of $A\Delta^H$ is $\eta_{j-1+k} = b^{-2} \sum_{l \geq k} y_l/(j+l)$ in the window $k_- \leq k \leq k_+$, and $\eta_n = \eta_{j-1+k_-}$ for $M < n < j - 1 + k_-$. Writing $\frac{y_l}{j+l} = \frac{y_l}{j+1}(1 + \frac{1-l}{j+1})$ and using $|\frac{1-l}{j+1}| \leq \frac{|l-1|}{j_*+k_-}$, we get $|\eta_{j-1+k} - \bar{\eta}_k| \leq \delta_k$, hence $|\eta_{j-1+k}| \leq E_k$, and $|\eta_n| \leq E_{k_-}$ below the window. We now bound the parts of $A\Delta^H$, divided by $\omega_j = b^2(j+1)\rho^j$.

- Shape coordinates in the window: $\omega_{j-1+k}/\omega_j = (1 + \frac{k-1}{j+1})\rho^{k-1}$, which gives at most $\mathbf{U}_1$.
- Shape coordinates below the window: $\sum_{n \leq j-3+k_-} (n+1)\rho^n/((j+1)\rho^j) \leq \rho^{k_- - 3}\tau$, which gives at most $\mathbf{U}_2$.
- The coupled coordinates $\delta g_{n-2,1} = b^2\eta_n$, of weight ρ^{n-2} : at most $\mathbf{U}_3$, by the same two computations and $j+1 > j_* + 1$.
- The finite part $b^2\eta_{M+2}\hat{A}((M+1)e_{(M,0)} + e_{(M,1)})$: its norm is at most $b^2HE_{k_-}$, which gives at most $\mathbf{U}_4$ since $j > j_*$.

The nonharmonic part. Δ^N lies in nonharmonic tail rows, on which A is the identity, and its normalised norm is at most

$$\sum_{\alpha,\beta} \frac{|jq_{\alpha\beta}^1 + q_{\alpha\beta}^0|}{j+1} \cdot \frac{\beta}{j+\alpha} \cdot \frac{\rho^{j-1+\alpha-\beta}}{b^2\rho^j} \leq \mathbf{U}_5.$$

The part Δ^W . We have $jw^{j-1}\bar{p}W^\circ = j(1 - |w|^2)^2w^{j-1-\Lambda}(w^\Lambda\bar{p}Q^\circ)$ with $j - 1 - \Lambda \geq 0$. Since conjugation preserves $\|\cdot\|_\rho$, $\|2\operatorname{Re} f\|_\rho \leq 2\|f\|_\rho$, and Lemma 8.3 with $N = j - 1 - \Lambda$ gives

$$\frac{\|\Delta^W\|_\rho}{\omega_j} \leq \frac{2j}{b^2(j+1)\rho^j} \sum_{m,t} |\varpi_{m,t}| \frac{4(2t+1)(2t+3)\rho^{m+j-1-\Lambda}}{(m+j-\Lambda)^2} \leq \mathbf{U}_W.$$

All frequencies in Δ^W are at least $j - 1 - \Lambda > M$ because $j_* > 2M + d + 2$, so $\|A\Delta^W\|_X \leq A_t\|\Delta^W\|_\rho$ by Lemma 7.6.

Adding the three contributions proves the claim. $\square$

Remark 8.5. Each of ε_k , δ_k , U_3 , U_5 is $O(1/j_*)$, $U_4 = O(\rho^{-j_*}/j_*)$ and $U_W = O(j_*^{-2})$. Hence, as $j_* \rightarrow \infty$, the majorant $Z_{\text{sh},\infty}$ tends to $\sum_k \rho^{k-1} |\bar{\eta}_k| + |\bar{\eta}_{k-}| \rho^{k-3} \tau \approx 0.19723$; for comparison, the directly computed normalised column at $j = 1003$ is about 0.19752.

Collecting the five classes, the number

$$Z = \max\{Z_f, Z_{g,\partial}, Z_{g,\infty}, Z_{\text{sh},\partial}, Z_{\text{sh},\infty}\}$$

bounds $\|I - ADF(x^\circ)\|_{X \rightarrow X}$.

9. CERTIFIED BOUNDS AND THE EXACT ZERO

The quantities defined in Sections 6–8 are finite expressions in the 525 dyadic centre coefficients and in $\rho = 21/20$, apart from the entries of $\hat{A}$, which are dyadic numbers produced by the computation and are then treated as exact data. They were evaluated in ball arithmetic as described in Section 13. Table 1 lists the results, rounded outward.

Quantity	Where defined	Certified bound
$b = c_1^\circ$	Section 6	14.8400430096918... (exact dyadic)
$P = N_\rho(p)$	Section 6	< 16.833325
$V_0 = \ V^\circ\ _\rho$	Section 6	< 134.261058
$\ I - \hat{A}M_f\ $	(7.5)	$< 1.343 \cdot 10^{-34}$
$\ R\ $	Lemma 7.6	< 675.571877
H	Lemma 7.6	< 1966.719238
A_t ($N_H = 443$)	Lemma 7.6	< 10.777973
$\ A\ $	Lemma 7.6	< 675.571877
Y	Section 8.1	$< 6.297631 \cdot 10^{-6}$
Z_f (525 columns)	Section 8.1	< 0.180814
$2P_s P_t + P_t^2$	Lemma 8.1	< 0.056966
$Z_{g,\partial}$ (3650 columns, maximum at (43, 1))	Lemma 8.1	< 0.313479
$Z_{g,\infty}$	Lemma 8.2	< 0.186417
$Z_{\text{sh},\partial}$ (480 columns, maximum at $j = 43$)	Section 8	< 0.360262
U_1	Proposition 8.4	< 0.217542
U_2	Proposition 8.4	$< 4.45704 \cdot 10^{-4}$
U_3	Proposition 8.4	$< 1.97149 \cdot 10^{-4}$
U_4	Proposition 8.4	$< 3.018 \cdot 10^{-24}$
U_5	Proposition 8.4	$< 9.23594 \cdot 10^{-4}$
$\sum_{m,t} \varpi_{m,t} (2t+1)(2t+3) \rho^{m-1-\Lambda}$	Proposition 8.4	< 19930.219512
$U_W, A_t U_W$	Proposition 8.4	$< 8.55375 \cdot 10^{-4}, < 9.219199 \cdot 10^{-3}$
$Z_{\text{sh},\infty}$ (all $j > 1001$)	Proposition 8.4	< 0.228328
Z	Section 8	< 0.360262
$C_2 = \ A\ _{c_2}$	Lemma 6.2	< 3.278617
$C_3 = \ A\ _{c_3}$	Lemma 6.2	$< 8.42289 \cdot 10^{-4}$

TABLE 1. Certified bounds, rounded outward, for $M = 41$, $S = 24$, $\rho = 21/20$, $j_* = 1001$. The complete interval enclosures are recorded in the certificate file (Section 13).

Theorem 9.1. *Let $r = 1/50000$. There is exactly one $x^* = (g^*, c^*) \in X$ with $\|x^* - x^\circ\|_X \leq r$ and $\mathcal{F}(x^*) = 0$.*

Proof. The conditions $j_* + k_- - 1 = 919 > M + 2$ and $j_* = 1001 > 2M + d + 2 = 124$ of Proposition 8.4 hold, and (7.5) holds by Table 1; hence A is bounded and injective (Lemma 7.5). By Table 1, Proposition 7.1 applies with the rational majorants $Y = 7 \cdot 10^{-6}$, $Z = 37/100$, $C_2 = 4$ and $C_3 = 10^{-3}$. For these values and $r = 1/50000$, exact rational arithmetic gives

$$Y + (Z - 1)r + C_2 r^2 + C_3 r^3 = -5.5983999999920 \cdot 10^{-6} < 0,$$

$$Z - 1 + 2C_2 r + 3C_3 r^2 = -0.6298399999988 < 0. \quad \square$$

Remark 9.2 (The centre). The centre x° was obtained numerically, and its provenance plays no role in the proof. A spectral collocation computation in spherical coordinates, started from the ball of radius $j_{2,4}$ perturbed by the zonal harmonic of degree six (see Section 14), produced an approximate solution of (1.2); its meridian domain was mapped conformally to the disc, and Newton's method was applied to the Galerkin truncation of (6.1) to the 21 shape coefficients $c_1, c_3, \dots, c_{41}$ and the 504 coefficients of g described above. The resulting binary64 numbers were then frozen and are interpreted as exact dyadic rationals.

10. RECONSTRUCTION OF THE DOMAIN FOR $n = 4$

Let $x^* = (g^*, c^*)$ be the zero of Theorem 9.1, $\psi^* = \psi_{c^*}$, and $\delta c = c^* - c^\circ$. Then $\sum_j \omega_j |\delta c_j| \leq r$, and consequently

$$(10.1) \quad \begin{aligned} |c_1^* - b| &\leq \frac{r}{2b^2\rho}, & \sum_j j\rho^{j-1} |\delta c_j| &\leq \frac{r}{b^2\rho}, \\ \sum_{j \geq 3} |\delta c_j| &\leq \frac{r}{4b^2\rho^3}, & \sum_j j(j-1) |\delta c_j| &\leq \frac{r\rho}{b^2(\rho-1)^2}, \end{aligned}$$

because $j\rho^{j-1}/\omega_j \leq 1/(b^2\rho)$, $1/\omega_j \leq 1/(4b^2\rho^3)$ for $j \geq 3$, and $j(j-1)/\omega_j \leq j\rho^{-j}/b^2 \leq \sum_{i \geq 1} i\rho^{-i}/b^2 = \rho/(b^2(\rho-1)^2)$. In particular ψ^* is holomorphic in $|w| < \rho$.

10.1. Univalence, star-shapedness and the non-disc property.

Proposition 10.1. *The map ψ^* is injective on the disc $\{|w| < 41/40\}$, and $\psi^{*'} has no zeros there. The domain $D = \psi^*(\mathbb{D})$ is bounded by the real-analytic Jordan curve $\psi^*(\partial\mathbb{D})$, it is strictly star-shaped with respect to 0, symmetric under $x_1 \mapsto -x_1$ and $y \mapsto -y$, and it is not a disc.$*

Proof. Let $R = 41/40 < \rho$, $b_- = b - r/(2b^2\rho) \leq c_1^*$, and

$$s_1 = \sum_{j \geq 3} j|c_j^\circ| + \frac{r}{b^2\rho}, \quad s_0 = \sum_{j \geq 3} |c_j^\circ| + \frac{r}{4b^2\rho^3}.$$

By (10.1), $\sum_{j \geq 3} j|c_j^*| \leq s_1$ and $\sum_{j \geq 3} |c_j^*| \leq s_0$. The computation (Section 13) certifies

$$(10.2) \quad \operatorname{Re} \psi^{*'}(w) \geq b_- - \sum_{j \geq 3} j|c_j^\circ| R^{j-1} - \frac{r}{b^2\rho} > 13.081358 \quad (|w| \leq R),$$

$$(10.3) \quad \frac{s_1}{b_-} < 0.104547, \quad \frac{s_1 + s_0}{b_- - s_0} < 0.126890, \quad |c_3^*| \geq |c_3^\circ| - \frac{r}{4b^2\rho^3} > 0.0911958,$$

where in (10.2) we used $j|\delta c_j| R^{j-1} \leq j|\delta c_j| \rho^{j-1}$ and (10.1).

Univalence. For $w_1 \neq w_2$ in the convex disc $|w| < R$, $\frac{\psi^*(w_2) - \psi^*(w_1)}{w_2 - w_1} = \int_0^1 \psi^{*'}(w_1 + \sigma(w_2 - w_1)) d\sigma$ has positive real part by (10.2).

Star-shapedness. For $|w| = 1$ we have $|\psi^*(w) - c_1^* w| \leq s_0$, $|w\psi^{*'}(w) - \psi^*(w)| \leq \sum_{j \geq 3} (j-1)|c_j^*| \leq s_1$ and $|\psi^*(w)| \geq b_- - s_0 > 0$, hence $|w\psi^{*'}/\psi^* - 1| < 0.127$ by (10.3). Therefore

$\frac{d}{d\theta} \arg \psi^*(e^{i\theta}) = \operatorname{Re} (w\psi^{*'}(w)/\psi^*(w)) > 0$. The winding number of $\psi^*(e^{i\theta})$ about 0 equals that of $c_1^*e^{i\theta}$, namely one, since $|\psi^*(w) - c_1^*w| \leq s_0 < c_1^*$ on $\partial\mathbb{D}$. Hence the argument increases by exactly 2π , and ∂D is a radial graph $\{R_D(\phi)e^{i\phi}\}$ with R_D positive and real analytic; D is strictly star-shaped.

Symmetry. ψ^* is odd with real coefficients, so $\psi^*(\bar{w}) = \overline{\psi^*(w)}$ and $\psi^*(-w) = -\psi^*(w)$.

Not a disc. If D were a disc, the two reflection symmetries would force its centre to be 0, and $w \mapsto \psi^*(w)/\psi^{*'}(0)$ would be a conformal map of $\mathbb{D}$ onto a disc centred at 0 fixing 0 with derivative one there; by the Schwarz lemma applied to it and to its inverse, it is the identity, so ψ^* is linear. This contradicts $c_3^* \neq 0$. $\square$

10.2. Convexity.

Proposition 10.2. *For $|w| \leq 1$,*

$$\operatorname{Re} \left(1 + \frac{w\psi^{*''}(w)}{\psi^{*'}(w)} \right) > 0.414074.$$

Consequently D is strictly convex, and $\Omega = \Pi^{-1}(D)$ is convex.

Proof. Let $A_1 = \sum_{j \geq 3} j|c_j^o|$, $A_2 = \sum_{j \geq 3} j(j-1)|c_j^o|$, $E_1 = r/(b^2\rho)$ and $E_2 = r\rho/(b^2(\rho-1)^2)$. By (10.1), for $|w| \leq 1$,

$$|\psi^{*'}(w)| \geq c_1^* - \sum_{j \geq 3} j|c_j^*| \geq b - A_1 - E_1 =: L, \quad |\psi^{*''}(w)| \leq \sum_{j \geq 3} j(j-1)|c_j^*| \leq A_2 + E_2 =: U.$$

The computation certifies $A_1 < 1.5514770$, $A_2 < 7.7860764$, $E_1 < 8.6491 \cdot 10^{-8}$, $E_2 < 3.81425 \cdot 10^{-5}$, hence $L > 13.288565$, $U < 7.786115$ and $\operatorname{Re}(1 + w\psi^{*''}/\psi^{*'}) \geq 1 - U/L > 0.414074$.

Let $z(\theta) = \psi^*(e^{i\theta})$. Then $z'(\theta) = ie^{i\theta}\psi^{*'}(e^{i\theta}) \neq 0$ and

$$\frac{d}{d\theta} \arg z'(\theta) = \operatorname{Re} \left(1 + \frac{w\psi^{*''}(w)}{\psi^{*'}(w)} \right) \Big|_{w=e^{i\theta}} > 0.$$

Thus the tangent direction of the simple closed analytic curve ∂D turns strictly monotonically, through a total angle 2π (the winding number of $\psi^{*'}(e^{i\theta})$ about 0 is zero, as $\psi^{*'}$ has no zeros in $\overline{\mathbb{D}}$), and ∂D bounds a strictly convex domain; this is the classical analytic convexity criterion for conformal maps of the disc, cf. [17, Chapter 2].

Now let $(x_1, x'), (z_1, z') \in \Omega$, $\sigma \in [0, 1]$, and put $y_\sigma = \sigma|x'| + (1-\sigma)|z'|$. By convexity of D , $(\sigma x_1 + (1-\sigma)z_1, \pm y_\sigma) \in D$, and by convexity again the vertical segment joining these two points lies in D . Since $|\sigma x' + (1-\sigma)z'| \leq y_\sigma$, the point $\sigma(x_1, x') + (1-\sigma)(z_1, z')$ lies in Ω . $\square$

10.3. Analytic regularity and continuation.

Lemma 10.3. *Let $U \subset \mathbb{R}^2$ be open and $F \in C(U)$ with $\int_U F(\Delta\varphi + \varphi) = 0$ for all $\varphi \in C_c^\infty(U)$. Then F is real analytic in U .*

Proof. Let $\mathcal{H}(x_1, y, \zeta) = e^\zeta F(x_1, y)$ on $U \times \mathbb{R}$. For $\Phi \in C_c^\infty(U \times \mathbb{R})$, integration by parts in ζ gives $\int \mathcal{H} \Delta_3 \Phi = \int_U F(\Delta\varphi + \varphi) = 0$ with $\varphi = \int e^\zeta \Phi(\cdot, \zeta) d\zeta \in C_c^\infty(U)$. By Weyl's lemma the continuous function $\mathcal{H}$ is harmonic, hence real analytic, and so is $F = \mathcal{H}(\cdot, \cdot, 0)$. $\square$

Lemma 10.4. *Let $D \subset \mathbb{R}^2$ be a bounded domain whose boundary is a real-analytic regular Jordan curve, and let $F \in C^1(\overline{D})$ satisfy $\int_D (\nabla F \cdot \nabla \varphi - F\varphi) = 0$ for all $\varphi \in C_c^\infty(D)$, together with $F = y$ and $\nabla F = e_y$ on ∂D . Then F extends to a real-analytic solution of $\Delta F + F = 0$ on an open neighbourhood of $\overline{D}$.*

Proof. Local Cauchy problem. Fix $z_0 \in \partial D$ and a real-analytic regular parametrisation γ of ∂D near z_0 with $\gamma(0) = z_0$; extend γ holomorphically to a disc $\{|a| < a_0\}$ on which $\gamma' \neq 0$ and γ is injective, and put $\gamma^\dagger(b) = \overline{\gamma(\bar{b})}$. In the complex coordinates $\zeta = x_1 + iy$, $\chi = x_1 - iy$ the Laplacian is $4\partial_\zeta\partial_\chi$, and the substitution $\zeta = \gamma(a)$, $\chi = \gamma^\dagger(b)$ turns $4\partial_\zeta\partial_\chi F + F = 0$ into $\partial_a\partial_b F = -kF$ with $k(a, b) = \gamma'(a)\gamma'^\dagger(b)/4$. The real plane corresponds to $b = \bar{a}$, and ∂D to the real diagonal $a = b \in \mathbb{R}$. The function $F_0(a, b) = (\gamma(a) - \gamma^\dagger(b))/(2i)$ is the holomorphic extension of y and satisfies $\partial_a\partial_b F_0 = 0$. On the closed bidisc $\{|a|, |b| \leq \varepsilon\}$ consider

$$(\mathcal{T}W)(a, b) = - \int_{[b, a]} \int_{[t, b]} k(t, v) (W(t, v) + F_0(t, v)) dv dt$$

along straight segments, which stay in the bidisc by convexity. $\mathcal{T}$ maps the Banach space of functions continuous on the closed bidisc and holomorphic in its interior into itself, and $\|\mathcal{T}W - \mathcal{T}\tilde{W}\|_\infty \leq 2\varepsilon^2 K_0 \|W - \tilde{W}\|_\infty$ with $K_0 = \sup |k|$, since $\int_{[b, a]} |t - b| |dt| = |a - b|^2/2 \leq 2\varepsilon^2$. For $2\varepsilon^2 K_0 < 1$ the contraction mapping principle gives a fixed point W , and differentiation of the iterated integral gives $W_{ab} = -k(W + F_0)$ and $W = W_a = W_b = 0$ on the diagonal $a = b$. Hence $\tilde{F}(x_1, y) = (F_0 + W)(\gamma^{-1}(x_1 + iy), (\gamma^\dagger)^{-1}(x_1 - iy))$ is a real-analytic (possibly complex-valued) solution of $\Delta\tilde{F} + \tilde{F} = 0$ on a disc U about z_0 , with $\tilde{F} = y$ and $\nabla\tilde{F} = e_y$ on $\partial D \cap U$.

Matching. Shrink U so that ∂D divides it into two connected pieces. The function $Z = F - \tilde{F}$ on $D \cap U$, extended by zero to $U \setminus D$, is C^1 on U , because its value and gradient vanish on $\partial D \cap U$. It is a weak solution of $\Delta Z + Z = 0$ on U . Indeed, let $\varphi \in C_c^\infty(U)$, and let $\chi_\delta \in C_c^\infty(D)$ be equal to one at distance at least 2δ from ∂D , with $0 \leq \chi_\delta \leq 1$ and $|\nabla\chi_\delta| \leq C/\delta$. Since F and $\tilde{F}$ are weak solutions in $D \cap U$ and $\chi_\delta\varphi \in C_c^\infty(D \cap U)$,

$$\int_U (\nabla Z \cdot \nabla \varphi - Z\varphi) = \int_{D \cap U} (1 - \chi_\delta) (\nabla Z \cdot \nabla \varphi - Z\varphi) - \int_{D \cap U} \varphi \nabla Z \cdot \nabla \chi_\delta.$$

The integrands are supported in the layer $\Sigma_\delta = \{\text{dist}(\cdot, \partial D) < 2\delta\} \cap D \cap \text{supp } \varphi$, whose area is $O(\delta)$. The first integral is $O(\delta)$, and the second is bounded by $C' \sup_{\Sigma_\delta} |\nabla Z|$, which tends to zero because ∇Z is continuous and vanishes on $\partial D \cap U$. By Lemma 10.3 (applied to the real and imaginary parts) Z is real analytic on U ; it vanishes on the open set $U \setminus \bar{D}$, hence on U . Thus $\tilde{F} = F$ on $D \cap U$.

Gluing. Cover ∂D by finitely many discs $B(z_i, r_i/2)$ such that on $B(z_i, r_i)$ a local extension $\tilde{F}_i$ as above exists, and let $0 < \varepsilon \leq \min_i r_i/6$. For z with $\text{dist}(z, \partial D) < \varepsilon$ choose $q \in \partial D$ with $|z - q| < \varepsilon$ and i with $q \in B(z_i, r_i/2)$, and put $\tilde{F}(z) = \tilde{F}_i(z)$. If another choice q', i' is made, then $|q - q'| < 2\varepsilon$, so $B(q', \varepsilon) \subset B(z_i, r_i) \cap B(z_{i'}, r_{i'})$; both $\tilde{F}_i$ and $\tilde{F}_{i'}$ coincide with F on the nonempty open set $B(q', \varepsilon) \cap D$, hence on the disc $B(q', \varepsilon) \ni z$. Thus $\tilde{F}$ is well defined and real analytic on $\{\text{dist}(\cdot, \partial D) < \varepsilon\}$, and together with F it gives the required extension. It is real valued because it is real on D . $\square$

10.4. Proof of Theorem 1.1 for $n = 4$. Let $V^* = Kg^* + \text{Im } \psi^*$. By Lemma 6.1, $V^* \in C^1(\bar{\mathbb{D}})$ satisfies (3.1) with $\psi = \psi^*$ and has the parities required in Lemma 3.1. By Proposition 10.1, ψ^* is injective with nonvanishing derivative on a neighbourhood of $\bar{\mathbb{D}}$. Lemma 3.1 shows that $F = V^* \circ (\psi^*)^{-1} \in C^1(\bar{D})$ is a weak solution of $\Delta F + F = 0$ in D with $F = y$ and $\nabla F = e_y$ on ∂D , odd in y and even in x_1 . By Lemmas 10.3 and 10.4, F extends to a real-analytic solution on an open neighbourhood of $\bar{D}$; replacing this neighbourhood by the connected component containing $\bar{D}$ of its intersection with its images under the two reflections, we obtain a symmetric connected neighbourhood N on which, by the identity theorem, F is still odd in y and even in x_1 .

Since D is strictly star-shaped and symmetric, ∂D meets the axis in exactly two points. Proposition 2.3 now shows that $\Omega = \Pi^{-1}(D)$ and $u = F(x_1, |x'|)/|x'|$ (extended analytically to the axis) satisfy (1.2), that u is real analytic on the neighbourhood $\Pi^{-1}(N)$ of $\bar{\Omega}$, and that u is nonconstant.

Geometry of Ω . Let $R_D(\phi)$ be the radial function of D from Proposition 10.1. It is positive, real analytic, 2π -periodic and even. Its Fourier series $R_D(\phi) = \sum_{k \geq 0} \alpha_k \cos k\phi$ has exponentially decaying coefficients, so $\tilde{R}(\tau) = \sum_k \alpha_k T_k(\tau)$, with T_k the Chebyshev polynomials, converges on a complex neighbourhood of $[-1, 1]$ (since $|T_k| \leq \varrho^k$ on the Bernstein ellipse of parameter ϱ) and defines a real-analytic function with $\tilde{R}(\cos \phi) = R_D(\phi)$, positive near $[-1, 1]$. For $x \neq 0$, $\Pi(x)$ has polar angle $\phi = \arccos(x_1/|x|) \in [0, \pi]$ and modulus $|x|$. Hence

$$\Omega = \{0\} \cup \{x \neq 0 : |x| < \tilde{R}(x_1/|x|)\}, \quad \partial\Omega = \{\tilde{R}(\omega_1)\omega : \omega \in S^3\}.$$

Thus Ω is strictly star-shaped with respect to 0, in particular connected. The map $\omega \mapsto \tilde{R}(\omega_1)\omega$ is real analytic and injective on S^3 (its inverse is $x \mapsto x/|x|$), and its differential maps a tangent vector $v \perp \omega$ to $\tilde{R}'(\omega_1)v_1\omega + \tilde{R}(\omega_1)v$, whose component tangent to S^3 is $\tilde{R}(\omega_1)v \neq 0$. Hence it is a real-analytic embedding, and $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^3 . The symmetries of Ω follow from $\Omega = \Pi^{-1}(D)$ and the symmetry of D under $x_1 \mapsto -x_1$.

Ω is not a ball. If it were, invariance under $x' \mapsto -x'$ and $x_1 \mapsto -x_1$ would put its centre at 0; then $\Pi(\Omega) = D \cap \{y \geq 0\}$ would be the upper half of a disc centred at 0, and by the symmetry $y \mapsto -y$, D would be that disc, contradicting Proposition 10.1. This completes the proof of Theorem 1.1 for $n = 4$, and Proposition 10.2 proves Corollary 1.3 for $n = 4$. $\square$

11. DIMENSION SIX

In this section we prove Theorem 1.1 and Corollary 1.3 for $n = 6$. The argument follows Sections 2–10 step by step. We state explicitly which results of those sections are used verbatim, and why, and we give complete proofs of everything that changes. Points of $\mathbb{R}^6$ are written $x = (x', x'') \in \mathbb{R}^3 \times \mathbb{R}^3$, points of the meridian plane $(y_1, y_2) \in \mathbb{R}^2$, identified with $z = y_1 + iy_2 \in \mathbb{C}$, and

$$\Pi_6 : \mathbb{R}^6 \rightarrow \mathbb{R}^2, \quad \Pi_6(x', x'') = (|x'|, |x''|).$$

The map Π_6 is continuous, its image is the closed quadrant $[0, \infty)^2$, and $|\Pi_6(x)| = |x|$.

11.1. Reduction to a planar problem.

Lemma 11.1. *Let f be a C^2 function on an open subset of $\{y_1 > 0, y_2 > 0\}$ and $F = y_1 y_2 f$. Then*

$$F_{y_1 y_1} + F_{y_2 y_2} = y_1 y_2 \left(f_{y_1 y_1} + f_{y_2 y_2} + \frac{2}{y_1} f_{y_1} + \frac{2}{y_2} f_{y_2} \right).$$

The expression in parentheses, evaluated at $\Pi_6(x)$, is the Laplacian of the function $x \mapsto f(|x'|, |x''|)$ at every point with $x' \neq 0$ and $x'' \neq 0$.

Proof. $(y_1 y_2 f)_{y_1 y_1} = y_2 (y_1 f_{y_1 y_1} + 2f_{y_1})$, and similarly for y_2 . The Laplacian of $\mathbb{R}^3 \times \mathbb{R}^3$ is the sum of the Laplacians of the two factors, each of which acts on functions of the modulus y of its variable as $\partial_y^2 + \frac{2}{y} \partial_y$. $\square$

Analyticity across the two axes is obtained by applying the division argument of Lemma 2.2 in each variable.

Lemma 11.2. *Let $N \subset \mathbb{R}^2$ be open and invariant under $(y_1, y_2) \mapsto (-y_1, y_2)$ and $(y_1, y_2) \mapsto (y_1, -y_2)$, and let F be real analytic on N , odd in y_1 and odd in y_2 . Then there is a real-analytic function G , defined on an open set containing $\{(y_1^2, y_2^2) : (y_1, y_2) \in N\}$, such that*

$F(y_1, y_2) = y_1 y_2 G(y_1^2, y_2^2)$ on N . Consequently $x \mapsto G(|x'|^2, |x''|^2)$ is real analytic on the open set $\Pi_6^{-1}(N)$, and it equals $F(\Pi_6(x))/(|x'| |x''|)$ where $x' \neq 0$ and $x'' \neq 0$.

Proof. Let $a = (a_1, a_2) \in N$. Choose $\delta > 0$, with $\delta < |a_i|$ whenever $a_i \neq 0$, such that $F(y) = \sum_{i,k \geq 0} c_{ik} (y_1 - a_1)^i (y_2 - a_2)^k$ converges absolutely for $|y_1 - a_1|, |y_2 - a_2| < \delta$. For $i = 1, 2$ let $I_i = (-\delta^2, \delta^2)$ if $a_i = 0$, and $I_i = (|a_i| - \delta)^2, (|a_i| + \delta)^2) \subset (0, \infty)$ if $a_i \neq 0$, and let σ_i be the sign of a_i . We define G on the box $I_1 \times I_2$.

- If $a_1 = a_2 = 0$, oddness in both variables gives $c_{ik} = 0$ unless i and k are odd, and we put $G(t_1, t_2) = \sum_{i,k} c_{2i+1, 2k+1} t_1^i t_2^k$.
- If $a_1 = 0 \neq a_2$, oddness in y_1 gives $c_{ik} = 0$ for even i , so $F = y_1 H(y_1^2, y_2)$ with $H(t_1, y_2) = \sum_{i,k} c_{2i+1, k} t_1^i (y_2 - a_2)^k$, and we put $G(t_1, t_2) = H(t_1, \sigma_2 \sqrt{t_2})/(\sigma_2 \sqrt{t_2})$. The case $a_2 = 0 \neq a_1$ is symmetric.
- If $a_1 a_2 \neq 0$, we put $G(t_1, t_2) = F(\sigma_1 \sqrt{t_1}, \sigma_2 \sqrt{t_2})/(\sigma_1 \sigma_2 \sqrt{t_1 t_2})$.

In each case G is real analytic on $I_1 \times I_2$, because $\sqrt{t_i}$ is real analytic and positive on I_i when $a_i \neq 0$. By the oddness of F in each variable and the invariance of N , $G(t_1, t_2) = F(\sqrt{t_1}, \sqrt{t_2})/\sqrt{t_1 t_2}$ at the points of the box with $t_1, t_2 > 0$. The intersection of two such boxes is a product of two intervals, each of which is contained in $(0, \infty)$ or symmetric about 0; if it is nonempty, it is connected and contains points with $t_1, t_2 > 0$, at which the two definitions agree. By the identity theorem they agree on the whole intersection, and G is well defined. It satisfies $F = y_1 y_2 G(y_1^2, y_2^2)$ near every point of N . The last assertion follows because $x \mapsto (|x'|^2, |x''|^2)$ is polynomial. $\square$

Proposition 11.3 (From the plane to $\mathbb{R}^6$). *Let $D \subset \mathbb{R}^2$ be a bounded domain, invariant under $(y_1, y_2) \mapsto (-y_1, y_2)$ and $(y_1, y_2) \mapsto (y_1, -y_2)$, whose boundary is a Jordan curve meeting each coordinate axis in exactly two points. Let F be real analytic on an open set $N \supset \overline{D}$ invariant under the same two reflections, odd in y_1 and in y_2 , and such that*

$$(11.1) \quad \Delta F + F = 0 \text{ in } D, \quad F = y_1 y_2 \text{ and } \nabla F = (y_2, y_1) \text{ on } \partial D.$$

Put $\Omega = \Pi_6^{-1}(D)$ and let u be the function $x \mapsto G(|x'|^2, |x''|^2)$ of Lemma 11.2. Then Ω is a bounded open set, u is real analytic on the neighbourhood $\Pi_6^{-1}(N)$ of $\overline{\Omega}$, u is nonconstant, and (1.2) holds.

Proof. We follow the proof of Proposition 2.3. The set Ω is open and bounded because Π_6 is continuous and proper; $\overline{\Omega} \subset \Pi_6^{-1}(\overline{D}) \subset \Pi_6^{-1}(N)$ and $\partial\Omega \subset \Pi_6^{-1}(\partial D)$. Let $E = \{x : x' \neq 0, x'' \neq 0\}$; its complement is the union of two three-dimensional subspaces. On $\Omega \cap E$ we have $u = f \circ \Pi_6$ with $f = F/(y_1 y_2)$, and Lemma 11.1 gives $|x'| |x''| (\Delta u + u)(x) = (\Delta F + F)(\Pi_6(x)) = 0$. Since $\Omega \cap E$ is dense in Ω and $\Delta u + u$ is continuous, $\Delta u + u = 0$ in Ω .

Let $x \in \partial\Omega \cap E$. Then $\Pi_6(x) \in \partial D$ lies in the open quadrant, and $f = F/(y_1 y_2) = 1$ there. From $F_{y_1} = y_2 f + y_1 y_2 f_{y_1} = y_2$ and the analogous identity for y_2 we get $f_{y_1} = f_{y_2} = 0$, hence $\nabla u(x) = (f_{y_1} x'/|x'|, f_{y_2} x''/|x''|) = 0$.

It remains to show that $\partial\Omega \cap E$ is dense in $\partial\Omega$. By the reflection symmetries, the two points of ∂D on the axis $\{y_1 = 0\}$ are $(0, \pm a)$ with $a > 0$. The reflection $y_1 \mapsto -y_1$ fixes them and maps $\{y_1 > 0\}$ to $\{y_1 < 0\}$, so it exchanges the two arcs of ∂D between them, and one of these arcs lies in $\{y_1 > 0\}$. Hence $(0, a)$ is a limit of points of ∂D in the open quadrant; the same holds for the points of ∂D on the other axis. Every point of $\Pi_6^{-1}(\partial D)$ lies outside Ω and is a limit of points of Ω , so $\Pi_6^{-1}(\partial D) = \partial\Omega$, and lifting the above sequences by Π_6 shows that $\partial\Omega \cap E$ is dense in $\partial\Omega$. By continuity $u = 1$ and $\nabla u = 0$ on $\partial\Omega$. Finally, a constant solution of $\Delta u + u = 0$ vanishes identically. $\square$

As in Proposition 2.4, averaging over $O(3) \times O(3)$ shows that the reduction is reversible in the analytic class. Let $y_1 + iy_2 = re^{i\phi}$. Since $y_1 y_2 = \frac{1}{2} r^2 \sin 2\phi$ and $\sin m\phi = \sin 2\phi U_{m/2-1}(\cos 2\phi)$ for even m , the planar Helmholtz solution $F = J_m(r) \sin m\phi$, with $m \equiv 2 \pmod{4}$, corresponds to the six-dimensional Helmholtz solution

$$u = 2r^{-2} J_m(r) U_{m/2-1}(\cos 2\phi), \quad \cos 2\phi = \frac{|x'|^2 - |x''|^2}{|x|^2}.$$

For $m = 2$ this is the radial solution $2r^{-2} J_2(r)$, whose derivative is $-2r^{-2} J_3(r)$; the Schiffer balls of $\mathbb{R}^6$ at eigenvalue one have radii $j_{3,k}$, the positive zeros of J_3 .

11.2. Symmetry sector and conformal formulation. For a real sequence $c = (c_j)_{j \equiv 1 \pmod{4}}$ put $\psi_c(w) = \sum_{j \equiv 1 \pmod{4}} c_j w^j$. Since $i^j = i$ for $j \equiv 1 \pmod{4}$,

$$\psi_c(\bar{w}) = \overline{\psi_c(w)}, \quad \psi_c(-\bar{w}) = -\overline{\psi_c(w)}, \quad \psi_c(i\bar{w}) = i \overline{\psi_c(w)}.$$

Hence, if ψ_c is injective on $\mathbb{D}$, then $D = \psi_c(\mathbb{D})$ is invariant under $z \mapsto \bar{z}$, $z \mapsto -\bar{z}$ and $z \mapsto i\bar{z}$, that is, under $y_2 \mapsto -y_2$, $y_1 \mapsto -y_1$ and the exchange $(y_1, y_2) \mapsto (y_2, y_1)$.

Lemma 11.4. *Let ψ be holomorphic and injective on a neighbourhood of $\bar{\mathbb{D}}$, with $\psi' \neq 0$ there, let $D = \psi(\mathbb{D})$ and $h_\psi = \text{Im}(\psi^2)/2$. Let $V \in C^1(\bar{\mathbb{D}})$ be real valued and satisfy*

$$(11.2) \quad \Delta V + |\psi'|^2 V = 0 \text{ in } \mathcal{D}'(\mathbb{D}), \quad V - h_\psi = 0 \text{ and } \nabla(V - h_\psi) = 0 \text{ on } \partial\mathbb{D}.$$

Then $F = V \circ \psi^{-1} \in C^1(\bar{D})$ satisfies $\int_D (\nabla F \cdot \nabla \varphi - F \varphi) dA = 0$ for all $\varphi \in C_c^\infty(D)$, and $F = y_1 y_2$, $\nabla F = (y_2, y_1)$ on ∂D . If moreover $\psi = \psi_c$ as above and $V(\bar{w}) = -V(w)$, $V(-\bar{w}) = -V(w)$, $V(i\bar{w}) = V(w)$, then F is odd in y_1 and in y_2 and invariant under $(y_1, y_2) \mapsto (y_2, y_1)$.

Proof. Since $y_1 y_2 = \text{Im}(z^2)/2$, we have $h_\psi = (y_1 y_2) \circ \psi$ and $V - h_\psi = (F - y_1 y_2) \circ \psi$. With this change, the proof of Lemma 3.1 applies verbatim, and the parities of F follow from those of V and the displayed symmetries of ψ_c . $\square$

For $n \equiv 2 \pmod{4}$, $n \geq 2$, and $s \geq 0$ let $S_{n,s} = \text{Im } \Phi_{n,s} = r^n P_s^{(0,n)}(2r^2 - 1) \sin n\theta$. Fix $\rho = 11/10$ in this section, and let $\mathcal{S}_6$ be the real Banach space of $f = \sum_{n \equiv 2 \pmod{4}, s \geq 0} f_{n,s} S_{n,s}$ with $\|f\|_\rho = \sum \rho^n |f_{n,s}| < \infty$, which is also its norm in $\mathcal{B}_\rho$. Since $\Phi_{m,s}(\bar{w}) = \Phi_{-m,s}(w)$, $\Phi_{m,s}(-\bar{w}) = (-1)^m \Phi_{-m,s}(w)$ and $\Phi_{m,s}(i\bar{w}) = i^m \Phi_{-m,s}(w)$, the real-valued elements of $\mathcal{B}_\rho$ with the three parities of Lemma 11.4 are exactly the elements of $\mathcal{S}_6$: the first two parities force m to be even, and the third one forces $i^m = -1$. Let $\mathcal{G}_6 = \{g \in \mathcal{S}_6 : g_{n,0} = 0 \text{ for all } n\}$, and define K on $\mathcal{G}_6$ by (5.1), now for $n \equiv 2 \pmod{4}$.

Lemma 11.5 (Results used verbatim). *The following statements hold, with the proofs given in the previous sections.*

- (a) *Lemmas 4.1, 4.2, 4.3 and 4.4, and the bounds (4.4). They are stated for all $m \in \mathbb{Z}$ and $s \geq 0$.*
- (b) *Lemma 5.1(a)–(c), for every $n \geq 1$ and $s \geq 1$. Its proof uses only the explicit coefficients $a_{s,k}^{(n)}$ of (4.2) and Lemma 4.1(a), and never the parity of n . The absolute column sum of (5.1) is $1/(q(q+2))$ with $q = n + 2s$, as in Lemma 5.1(d); in the present sector $q \geq 4$, so $K: \mathcal{G}_6 \rightarrow \mathcal{S}_6$ is bounded with $\|K\| = 1/24$, attained at $(n, s) = (2, 1)$.*
- (c) *Lemma 5.2 and Remark 5.3, with $\mathcal{G}_6$, $\mathcal{S}_6$ and the harmonic polynomials $r^n \sin n\theta$, $n \equiv 2 \pmod{4}$ (and with $1/24$ in place of $1/15$ in the uniform bound).*
- (d) *Lemma 7.2, Proposition 7.1 in the quartic form of Proposition 11.8 below, and Lemma 7.5 for the operator A defined in (11.13) below; the proof of Lemma 7.5 uses only the block structure of A .*

- (e) Lemma 8.3, which is stated for all $m, t, N \geq 0$, and the connection formulas of Section 8 for the Jacobi polynomials $P_k^{(1,n)}$ and $P_k^{(2,n)}$, which hold for every $n \geq 0$.
- (f) Lemma 10.3, and Lemma 10.4 with the Cauchy data (y, e_y) replaced by $(y_1 y_2, (y_2, y_1))$. The only property of the datum used in the proof of Lemma 10.4 is that it is a harmonic polynomial: in the coordinates $\zeta = y_1 + i y_2$, $\chi = y_1 - i y_2$ one has $y_1 y_2 = (\zeta^2 - \chi^2)/(4i)$, whose holomorphic extension $F_0(a, b) = (\gamma(a)^2 - \gamma^\dagger(b)^2)/(4i)$ again satisfies $\partial_a \partial_b F_0 = 0$, and the rest of the proof is unchanged.

11.3. The quartic equation. In the rest of this section $\rho = 11/10$, $M = 58$ and $S = 30$. The centre $x^\circ = (g^\circ, c^\circ)$ consists of 450 coefficients $g_{n,s}^\circ$, $n \in \{2, 6, \dots, 58\}$, $1 \leq s \leq 30$, and 15 coefficients c_j° , $j \in \{1, 5, \dots, 57\}$; all other coefficients of x° vanish. Each of these 465 numbers is an explicitly given dyadic rational (a binary64 number interpreted exactly), listed in the ancillary file `center_r6.json`. We write $b = c_1^\circ \approx 28.864751$, $\alpha = b^3/2$ and

$$\omega_j = \alpha(j+2)\rho^{j+1} \quad (j \equiv 1 \pmod{4}).$$

Let X_6 be the real Banach space of pairs $x = (g, c)$, with $g \in \mathcal{G}_6$ and $c = (c_j)_{j \equiv 1 \pmod{4}}$ real, normed by $\|(g, c)\|_X = \|g\|_\rho + \|c\|$, $\|c\| = \sum_j \omega_j |c_j|$. With

$$\theta_0 = \frac{1}{3\alpha\rho}, \quad \theta_1 = \frac{1}{\alpha\rho^2},$$

we have $N_\rho(\psi_c) \leq \theta_0 \|c\|$ and $N_\rho(\psi'_c) \leq \theta_1 \|c\|$, because $\rho^j/\omega_j = 1/(\alpha(j+2)\rho)$ and $j\rho^{j-1}/\omega_j = j/(\alpha(j+2)\rho^2)$. In particular ψ_c is holomorphic in $|w| < \rho$ and continuous up to $|w| = \rho$. Put

$$h_c = \frac{1}{2} \operatorname{Im}(\psi_c^2) = \frac{1}{2} \sum_{j,k} c_j c_k S_{j+k,0} \in \mathcal{S}_6$$

(note that $j+k \equiv 2 \pmod{4}$), and define

$$(11.3) \quad \mathcal{F}_6(g, c) = g + |\psi'_c|^2 (Kg + h_c).$$

Lemma 11.6. $\mathcal{F}_6$ is a continuous polynomial map of degree four from X_6 to $\mathcal{S}_6$. If $\mathcal{F}_6(g, c) = 0$, then $V = Kg + h_c$ belongs to $C^1(\overline{\mathbb{D}})$, has the three parities of Lemma 11.4, and satisfies (11.2) with $\psi = \psi_c$.

Proof. $\psi'_c = \sum_j j c_j w^{j-1}$ has real coefficients and only powers divisible by four. Hence $|\psi'_c|^2$ is real and invariant under $w \mapsto \bar{w}$ and $w \mapsto iw$, multiplication by it preserves the three parities, and by (4.4) it maps $\mathcal{S}_6$ into $\mathcal{S}_6$ with norm at most $N_\rho(\psi'_c)^2$. Since $w^k = \Phi_{k,0}$ has norm ρ^k and $\|\operatorname{Im} f\|_\rho \leq \|f\|_\rho$, we have $\|h_c\|_\rho \leq N_\rho(\psi_c)^2/2$. Together with $\|K\| = 1/24$, this shows that (11.3) is a sum of bounded multilinear maps of degree at most four. The rest of the proof is that of Lemma 6.1, using Lemma 11.5(c) and the fact that h_c is harmonic. $\square$

11.4. Expansion at the centre. Write $p = \psi'_{c^\circ} = \sum_{a \in \{0,4,\dots,56\}} p_a w^a$ with $p_a = (a+1)c_{a+1}^\circ$, so $p_0 = b$ and $d = \deg p = 56$, and put

$$P = N_\rho(p), \quad C = N_\rho(\psi_{c^\circ}), \quad V^\circ = Kg^\circ + h_{c^\circ}, \quad V_0 = \|V^\circ\|_\rho.$$

For $h = (\delta g, \eta) \in X_6$ let

$$L(\eta) = 2 \operatorname{Re}(\bar{p} \psi'_\eta), \quad Q(\eta_1, \eta_2) = \operatorname{Re}(\psi'_{\eta_1} \overline{\psi'_{\eta_2}}), \quad \ell(\eta) = \operatorname{Im}(\psi_{c^\circ} \psi_\eta), \quad I(\eta_1, \eta_2) = \operatorname{Im}(\psi_{\eta_1} \psi_{\eta_2}),$$

and $\delta V = K\delta g + \ell(\eta)$. Since $|p + \psi'_\eta|^2 = |p|^2 + L(\eta) + Q(\eta, \eta)$ and $h_{c^\circ + \eta} = h_{c^\circ} + \ell(\eta) + \frac{1}{2}I(\eta, \eta)$, expanding (11.3) gives

$$(11.4) \quad \mathcal{F}_6(x^\circ + h) = \mathcal{F}_6(x^\circ) + D\mathcal{F}_6(x^\circ)h + B_2(h, h) + B_3(h, h, h) + B_4(h, h, h, h),$$

where

$$(11.5) \quad \begin{aligned} D\mathcal{F}_6(x^\circ)h &= \delta g + |p|^2(K\delta g + \ell(\eta)) + L(\eta)V^\circ, \\ B_2(h, h) &= \tfrac{1}{2}|p|^2 I(\eta, \eta) + L(\eta)\delta V + Q(\eta, \eta)V^\circ, \\ B_3(h, h, h) &= \tfrac{1}{2}L(\eta)I(\eta, \eta) + Q(\eta, \eta)\delta V, \quad B_4(h, h, h, h) = \tfrac{1}{2}Q(\eta, \eta)I(\eta, \eta), \end{aligned}$$

and B_2, B_3, B_4 denote the symmetric multilinear maps with these diagonals.

Lemma 11.7. *Let $\kappa = 1/24$. For all $h_i \in X_6$ and $k = 2, 3, 4$, $\|B_k(h_1, \dots, h_k)\|_\rho \leq c_k \|h_1\|_X \cdots \|h_k\|_X$, where*

$$c_2 = \max\{2P\theta_1\kappa, 2P\theta_1 C\theta_0 + \tfrac{1}{2}P^2\theta_0^2 + \theta_1^2 V_0\}, \quad c_3 = \max\{\theta_1^2\kappa, P\theta_1\theta_0^2 + \theta_1^2 C\theta_0\}, \quad c_4 = \tfrac{1}{2}\theta_1^2\theta_0^2.$$

Proof. Write $a_i = \|\delta g_i\|_\rho$ and $e_i = \|\eta_i\|$. As in the proof of Lemma 6.2, $N_\rho(\psi'_{\eta_i}) \leq \theta_1 e_i$, $N_\rho(\psi_{\eta_i}) \leq \theta_0 e_i$, $\|\ell(\eta_i)\|_\rho \leq C\theta_0 e_i$, $\|I(\eta_i, \eta_k)\|_\rho \leq \theta_0^2 e_i e_k$, $\|\delta V_i\|_\rho \leq \kappa a_i + C\theta_0 e_i$, $\|L(\eta_i)f\|_\rho \leq 2P\theta_1 e_i \|f\|_\rho$ and $\|Q(\eta_i, \eta_k)f\|_\rho \leq \theta_1^2 e_i e_k \|f\|_\rho$. Hence

$$\|B_2(h_1, h_2)\|_\rho \leq P\theta_1\kappa(e_1 a_2 + e_2 a_1) + (\tfrac{1}{2}P^2\theta_0^2 + 2P\theta_1 C\theta_0 + \theta_1^2 V_0)e_1 e_2 \leq c_2(a_1 + e_1)(a_2 + e_2).$$

In the bound for B_3 , every monomial $e_i e_k a_l$ has coefficient $\theta_1^2\kappa/3$ and $e_1 e_2 e_3$ has coefficient $P\theta_1\theta_0^2 + \theta_1^2 C\theta_0$; in the bound for B_4 , $e_1 e_2 e_3 e_4$ has coefficient $\tfrac{1}{2}\theta_1^2\theta_0^2$. Each is at most its coefficient in $c_k \prod_i (a_i + e_i)$. $\square$

Proposition 11.8. *Let $\mathcal{F}_6: X_6 \rightarrow \mathcal{S}_6$ have the form (11.4) with symmetric bounded B_2, B_3, B_4 , and let $A: \mathcal{S}_6 \rightarrow X_6$ be bounded, linear and injective. Suppose that $\|A\mathcal{F}_6(x^\circ)\|_X \leq Y$, $\|I - A D\mathcal{F}_6(x^\circ)\|_{X \rightarrow X} \leq Z$ and $\|A\| c_k \leq C_k$ for $k = 2, 3, 4$, and that $r > 0$ satisfies*

$$(11.6) \quad Y + (Z - 1)r + C_2 r^2 + C_3 r^3 + C_4 r^4 < 0, \quad Z - 1 + 2C_2 r + 3C_3 r^2 + 4C_4 r^3 < 0.$$

Then $\mathcal{F}_6$ has exactly one zero in the closed ball $\overline{B}_{X_6}(x^\circ, r)$.

Proof. This is the proof of Proposition 7.1, with one more term: the difference $B_4(h_1, \dots, h_1) - B_4(h_2, \dots, h_2)$ is a sum of four terms, each of which contains $h_1 - h_2$ once and three factors h_1 or h_2 , so that the norm of its image under A is at most $4C_4 r^3 \|h_1 - h_2\|_X$. $\square$

11.5. The principal part of the linearisation. In the model on high shape modes the map is replaced by the disc $\psi = bw$ and g by 0. We need the following radial expansion; here $S_{m,s} = r^m P_s^{(0,m)}(2r^2 - 1) \sin m\theta$ for any integer $m \geq 0$.

Lemma 11.9. *For every integer $m \geq 0$,*

$$r^{m+4} \sin m\theta = \frac{m+1}{m+3} S_{m,0} + \frac{2}{m+4} S_{m,1} + \frac{2}{(m+3)(m+4)} S_{m,2}.$$

Proof. Dividing by $r^m \sin m\theta$ and writing $t = r^2 = 1 + u$, the claim is $t^2 = \frac{m+1}{m+3} P_0 + \frac{2}{m+4} P_1 + \frac{2}{(m+3)(m+4)} P_2$ with $P_s = P_s^{(0,m)}(2t - 1)$. By (4.2), $P_0 = 1$, $P_1 = 1 + (m+2)u$ and $P_2 = 1 + 2(m+3)u + \frac{1}{2}(m+3)(m+4)u^2$. The right-hand side is therefore $c_0 + c_1 u + c_2 u^2$ with $c_2 = 1$, $c_1 = \frac{2(m+2)}{m+4} + \frac{4}{m+4} = 2$ and $c_0 = \frac{(m+1)(m+4)+2(m+3)+2}{(m+3)(m+4)} = 1$, which is $(1+u)^2 = t^2$. $\square$

Proposition 11.10. *For every $x = (g, c) \in X_6$ and every $j \equiv 1 \pmod{4}$,*

$$(11.7) \quad D\mathcal{F}_6(x)e_j = 2 \operatorname{Re}(jw^{j-1}\overline{\psi'_c})(Kg + h_c) + |\psi'_c|^2 \operatorname{Im}(\psi_c w^j).$$

Let $x_b = (0, b e_1)$, so that $\psi_{x_b}(w) = bw$. For every $j \equiv 1 \pmod{4}$ with $j \geq 5$,

$$(11.8) \quad T_j := D\mathcal{F}_6(x_b)e_j = b^3 \left\{ \frac{j+2}{2} S_{j+1,0} - \frac{j-2}{2} S_{j-3,0} - \frac{j}{j+1} S_{j-3,1} - \frac{1}{j+1} S_{j-3,2} \right\}.$$

Proof. Replacing c by $c + \varepsilon e_j$ changes ψ_c by εw^j ; since $\partial_\varepsilon |\psi'_c + \varepsilon j w^{j-1}|^2 = 2 \operatorname{Re}(j w^{j-1} \overline{\psi'_c})$ and $\partial_\varepsilon \frac{1}{2} \operatorname{Im}((\psi_c + \varepsilon w^j)^2) = \operatorname{Im}(\psi_c w^j)$ at $\varepsilon = 0$, we obtain (11.7). At x_b we have $g = 0$, $\psi'_c = b$ and $h_c = \frac{1}{2} b^2 \operatorname{Im}(w^2)$, so

$$T_j = b^3 [j \operatorname{Re}(w^{j-1}) \operatorname{Im}(w^2) + \operatorname{Im}(w^{j+1})] = b^3 [j r^{j+1} \cos((j-1)\theta) \sin 2\theta + r^{j+1} \sin((j+1)\theta)].$$

Since $2 \cos((j-1)\theta) \sin 2\theta = \sin((j+1)\theta) - \sin((j-3)\theta)$,

$$T_j = b^3 \left[\frac{j+2}{2} r^{j+1} \sin((j+1)\theta) - \frac{j}{2} r^{j+1} \sin((j-3)\theta) \right].$$

The first term is $\frac{j+2}{2} S_{j+1,0}$. Lemma 11.9 with $m = j-3 \geq 2$ gives

$$-\frac{j}{2} r^{j+1} \sin((j-3)\theta) = -\frac{j}{2} \left[\frac{j-2}{j} S_{j-3,0} + \frac{2}{j+1} S_{j-3,1} + \frac{2}{j(j+1)} S_{j-3,2} \right],$$

which is the rest of (11.8). $\square$

Both frequencies $j+1$ and $j-3$ in (11.8) are $\equiv 2 \pmod{4}$, so $T_j \in \mathcal{S}_6$. The harmonic part of T_j is $\alpha(j+2)$ in row $(j+1, 0)$ and $-\alpha(j-2)$ in row $(j-3, 0)$; this replaces the upper bidiagonal structure of Lemma 7.3 by a difference operator of step four.

11.6. Tail model and approximate inverse. The *finite inputs* are the 450 coordinates $g_{n,s}$ with $n \leq M$, $1 \leq s \leq S$, and the 15 shape coordinates c_j with $j \leq M-1$. The *finite rows* are the 450 rows (n, s) with $n \leq M$, $1 \leq s \leq S$, and the 15 harmonic rows $(n, 0)$ with $n \leq M$. All other inputs and rows are *tail* inputs and rows, and we write $X_6 = X_f \oplus X_t$, $\mathcal{S}_6 = \mathcal{S}_f \oplus \mathcal{S}_t$; $\dim X_f = \dim \mathcal{S}_f = 465$. Let M_f be the 465×465 matrix of the restriction of $D\mathcal{F}_6(x^\circ)$ to finite inputs and finite rows. On tail inputs we use the model

$$(11.9) \quad T(\delta g, \eta) = \delta g + \sum_{j \geq M+3} \eta_j T_j.$$

Write $Te = (Be, T_{tt}e)$ for tail inputs e , with B the part in finite rows. By (11.8) with $j = M+3 = 61$, only the shape input $j = M+3$ reaches finite rows:

$$(11.10) \quad B(\delta g, \eta) = -\eta_{M+3} \left(\alpha(M+1) e_{(M,0)} + \frac{b^3(M+3)}{M+4} e_{(M,1)} + \frac{b^3}{M+4} e_{(M,2)} \right).$$

Put $\tau_4 = (1 - \rho^{-4})^{-1}$.

Lemma 11.11. $T_{tt}: X_t \rightarrow \mathcal{S}_t$ is a bounded bijection with $\|T_{tt}\| \leq 1 + \rho^{-4}$. For a tail residual $y = \sum y_{n,s} S_{n,s} \in \mathcal{S}_t$ its inverse is given by

$$(11.11) \quad \eta_{n-1} = \sum_{k \geq 0} \frac{y_{n+4k,0}}{\alpha(n+4k+1)} \quad (n \geq M+4, n \equiv 2 \pmod{4}),$$

$$\delta g_{n,1} = y_{n,1} + \frac{b^3(n+3)}{n+4} \eta_{n+3}, \quad \delta g_{n,2} = y_{n,2} + \frac{b^3}{n+4} \eta_{n+3} \quad (n \geq M+4),$$

and $\delta g_{n,s} = y_{n,s}$ for all other tail g -inputs.

Proof. By (11.9) and (11.8), the tail harmonic rows read $\alpha(n+1)(\eta_{n-1} - \eta_{n+3}) = y_{n,0}$ for $n \geq M+4$, the tail rows $(n, 1)$ and $(n, 2)$ with $n \geq M+4$ read $\delta g_{n,1} - \frac{b^3(n+3)}{n+4} \eta_{n+3} = y_{n,1}$ and $\delta g_{n,2} - \frac{b^3}{n+4} \eta_{n+3} = y_{n,2}$, and all other tail rows read $\delta g_{n,s} = y_{n,s}$. Since $\sum \omega_j |\eta_j| < \infty$ forces $\eta_j \rightarrow 0$, summation of the harmonic equations gives (11.11), and a solution with $y = 0$ vanishes; conversely (11.11) solves the equations whenever the series converge, which is the case for every $y \in \mathcal{S}_t$ by the proof of Lemma 11.12 below (with $H = 0$). For $j \geq M+7$ the column $T_{tt}e_j$ has the four entries $\alpha(j+2)$, $-\alpha(j-2)$, $-b^3 j/(j+1)$ and $-b^3/(j+1)$ in

the rows $(j+1, 0)$, $(j-3, 0)$, $(j-3, 1)$ and $(j-3, 2)$; its weighted norm divided by ω_j is $1 + j\rho^{-4}/(j+2) \leq 1 + \rho^{-4}$. For $j = M+3$ only the first entry lies in a tail row, and tail g -inputs are mapped to the identical rows. Lemma 7.2 gives the bound for $\|T_{tt}\|$. $\square$

Let $\hat{A}$ be a 465×465 matrix with dyadic rational entries, regarded as an exact linear map $\mathcal{S}_f \rightarrow X_f$, such that

$$(11.12) \quad \|I - \hat{A}M_f\|_{X_f \rightarrow X_f} < 1,$$

and define

$$(11.13) \quad A = \begin{pmatrix} \hat{A} & -\hat{A}BT_{tt}^{-1} \\ 0 & T_{tt}^{-1} \end{pmatrix} : \mathcal{S}_f \oplus \mathcal{S}_t \rightarrow X_f \oplus X_t.$$

By Lemma 7.5 (Lemma 11.5(d)), A is bounded and injective, and $ATe = e$ for every tail input e . In the computation, $\hat{A}$ is a fixed binary64 approximation of M_f^{-1} , stored in the ancillary file `inverse_r6_M58_S30.npy`, whose entries are interpreted as exact dyadic rationals; only (11.12) is used.

Lemma 11.12. *Let*

$$\|R\| = \max_{(n,s) \text{ finite row}} \frac{\|\hat{A}e_{(n,s)}\|_X}{\rho^n},$$

$$H = \left\| \hat{A} \left(\alpha(M+1)e_{(M,0)} + \frac{b^3(M+3)}{M+4}e_{(M,1)} + \frac{b^3}{M+4}e_{(M,2)} \right) \right\|_X,$$

and, for $n \geq M+4$ with $n \equiv 2 \pmod{4}$,

$$(11.14) \quad h_n = \frac{1}{(n+1)\rho^n} \left(\sum_{\substack{M+4 \leq k \leq n \\ k \equiv 2(4)}} (k+1)\rho^k + 2 \sum_{\substack{M+4 \leq k \leq n-4 \\ k \equiv 2(4)}} \rho^k + \frac{H}{\alpha} \right).$$

Let $N_H \geq M+4$ with $N_H \equiv 2 \pmod{4}$. Then A maps every residual supported in tail rows with norm at most

$$A_t := \max \left\{ 1, \max_{M+4 \leq n \leq N_H} h_n, \tau_4 + \frac{2\tau_4\rho^{-4}}{N_H+5} + \frac{H}{\alpha(N_H+5)\rho^{N_H+4}} \right\},$$

it is the identity on residuals supported in nonharmonic tail rows, and $\|A\| \leq \max\{\|R\|, A_t\}$.

Proof. We use Lemma 7.2. A finite row $e_{(n,s)}$ is mapped to $\hat{A}e_{(n,s)}$. A nonharmonic tail row is mapped to the identical tail input by Lemma 11.11, and B vanishes on it. The normalised harmonic tail row $\rho^{-n}S_{n,0}$ is mapped to the shape coordinates $\eta_{k-1} = \rho^{-n}/(\alpha(n+1))$, $M+4 \leq k \leq n$, of weights $\omega_{k-1} = \alpha(k+1)\rho^k$; to the coordinates $\delta g_{k,1}$ and $\delta g_{k,2}$, $M+4 \leq k \leq n-4$, of weight ρ^k and total absolute value $b^3\rho^{-n}/(\alpha(n+1)) = 2\rho^{-n}/(n+1)$; and, by (11.10) and (11.13), to the finite part $\eta_{M+3}\hat{A}(\alpha(M+1)e_{(M,0)} + \dots)$, of norm $H\rho^{-n}/(\alpha(n+1))$. The sum of the three norms is exactly h_n . For $n \geq N_H+4$ we bound $(k+1)/(n+1) \leq 1$ in the first sum and sum the geometric series $\sum_{i \geq 0} \rho^{-4i} = \tau_4$; the second term is at most $2\tau_4\rho^{-4}/(n+1)$, and the third one decreases in n . $\square$

11.7. The defect bounds. We bound $\|I - A\mathcal{F}_6(x^\circ)\|$ as in Section 8, by partitioning the inputs into five classes; every input belongs to exactly one class, and no column is estimated by sampling.

The residual and the finite columns. $\mathcal{F}_6(x^\circ)$ is a polynomial in $w, \bar{w}$; its expansion in the basis $S_{n,s}$ (it has 2601 coefficients) is computed without truncation in interval arithmetic, using (4.3) and (5.1). Only finitely many tail rows occur, so $A\mathcal{F}_6(x^\circ)$ is computed without truncation

from (11.13), (11.10) and Lemma 11.11, and Y is an upper bound of its norm. For each of the 465 finite inputs e , $(I - AD\mathcal{F}_6(x^\circ))e$ is computed in the same way, and Z_f bounds the maximum of the normalised norms of these columns.

Tail g -columns.

Lemma 11.13. *Let $e = e_{(n,s)}$ be a tail g -input and $q = n + 2s$. Then $(I - AD\mathcal{F}_6(x^\circ))e = -A(|p|^2 K S_{n,s})$. Let $p = p_{\leq 24} + p_{> 24}$ be the splitting of p into terms of degree ≤ 24 and > 24 , and $P_s = N_\rho(p_{\leq 24})$, $P_t = N_\rho(p_{> 24})$, $\delta_a = 2P_s P_t + P_t^2$. Then*

$$(11.15) \quad \frac{\|(I - AD\mathcal{F}_6(x^\circ))e\|_X}{\rho^n} \leq \frac{\|A(|p_{\leq 24}|^2 K S_{n,s})\|_X}{\rho^n} + \frac{\|A\| \delta_a}{q(q+2)}.$$

Moreover, the right-hand side is at most $(A_t P_s^2 + \|A\| \delta_a)/(88 \cdot 90)$ if $n \geq M + 28 = 86$, and at most $(P_s^2 + \|A\| \delta_a)/(114 \cdot 116)$ if $s \geq S + 26 = 56$.

Proof. By (11.5), $D\mathcal{F}_6(x^\circ)e = S_{n,s} + |p|^2 K S_{n,s}$, and $Te = S_{n,s}$, so $AS_{n,s} = e$. The splitting is estimated as in Lemma 8.1. The function $|p_{\leq 24}|^2 = \sum_{a,a' \leq 24} p_a p_{a'} w^a \bar{w}^{a'}$ changes angular frequencies by at most 24 in modulus, and $\| |p_{\leq 24}|^2 K S_{n,s} \|_\rho \leq P_s^2 \rho^n / (q(q+2))$. If $n \geq 86$, all components of $|p_{\leq 24}|^2 K S_{n,s}$ have frequencies of modulus at least $n - 24 \geq 62 > M$, so they lie in tail rows, on which A has norm at most A_t (Lemma 11.12); and $q \geq 88$. If $s \geq 56$, then by (5.1) and Lemma 4.3 all components have radial index at least $s - 1 - 24 \geq 31 > S$, so they lie in nonharmonic tail rows, on which A is the identity; and $q \geq 2 + 2 \cdot 56 = 114$. $\square$

The tail g -inputs with $n \leq 82$ and $s \leq 55$ are the $21 \cdot 55 - 450 = 705$ near ones; for them the first term of (11.15) is computed without truncation, and $Z_{g,\partial}$ bounds the maximum of the right-hand side. The remaining tail g -inputs are covered by the last statement of Lemma 11.13; let $Z_{g,\infty}$ be the larger of the two bounds.

Near shape columns. For the 85 shape inputs e_j with $M + 3 = 61 \leq j \leq 397$, we have $(I - AD\mathcal{F}_6(x^\circ))e_j = -A\Delta_j$ with $\Delta_j = D\mathcal{F}_6(x^\circ)e_j - T_j$, because $ATe_j = e_j$; Δ_j is a polynomial, and these columns are computed without truncation. Let $Z_{\text{sh},\partial}$ bound their normalised norms.

Far shape columns. Let $W^\circ = Kg^\circ$ and $h^\circ = h_{c^\circ}$. By (11.7) at x° and at x_b , for every $j \equiv 1 \pmod{4}$, $j \geq 5$,

$$(11.16) \quad \Delta_j = 2 \operatorname{Re}(jw^{j-1}\Gamma_1) + \operatorname{Im}(w^j\Gamma_0) + 2 \operatorname{Re}(jw^{j-1}\bar{p}W^\circ),$$

where

$$\Gamma_1 = \bar{p}h^\circ - \frac{1}{2}b^3 \operatorname{Im}(w^2), \quad \Gamma_0 = |p|^2 \psi_{c^\circ} - b^3 w.$$

By Lemma 5.1(c) (Lemma 11.5(b)), $W^\circ = (1 - |w|^2)^2 Q^\circ$ with

$$Q^\circ = \sum_{n,s} \frac{g_{n,s}^\circ}{4s(s+1)} r^n P_{s-1}^{(2,n)} (2r^2 - 1) \sin n\theta,$$

and the connection formulas of Section 8 give the finite expansion of Q° in the basis $S_{n,t}$; its connection coefficients are positive rational numbers. Let $\Lambda = M + d = 114$. The function $w^\Lambda \bar{p} Q^\circ$ has only nonnegative frequencies; write $w^\Lambda \bar{p} Q^\circ = \sum_{m \geq 0, t \geq 0} \varpi_{m,t} \Phi_{m,t}$ (a finite sum, computed in interval arithmetic). Let $j_* = 401$ and

$$\mathsf{U}_\Gamma = \frac{1}{\alpha} \left(\frac{2\|\Gamma_1\|_\rho}{\rho^2} + \frac{\|\Gamma_0\|_\rho}{\rho(j_* + 2)} \right), \quad \mathsf{U}_W = \frac{8}{\alpha(j_* - \Lambda)^2} \sum_{m,t} |\varpi_{m,t}| (2t+1)(2t+3) \rho^{m-\Lambda-2},$$

where $\|\cdot\|_\rho$ is the norm of $\mathcal{B}_\rho$.

Proposition 11.14. *For every $j \equiv 1 \pmod{4}$ with $j \geq j_*$,*

$$\frac{\|(I - AD\mathcal{F}_6(x^\circ))e_j\|_X}{\omega_j} \leq Z_{\text{sh},\infty} := A_t(\mathbf{U}_\Gamma + \mathbf{U}_W).$$

Proof. The frequencies of $\bar{p}$ lie in $[-d, 0] = [-56, 0]$, those of h° in $[-114, 114]$, since h° is a combination of $S_{j+k,0}$ with $j, k \leq 57$, and those of W° and Q° in $[-M, M]$. Hence all frequencies of $w^{j-1}\Gamma_1$, $w^j\Gamma_0$ and $w^{j-1}\bar{p}W^\circ$ are at least $j-1-170 \geq 230 > M$. Taking real and imaginary parts only changes the signs of frequencies, so all components of $\Delta_j \in \mathcal{S}_6$ lie in tail rows, and $\|A\Delta_j\|_X \leq A_t\|\Delta_j\|_\rho$ by Lemma 11.12. By Lemma 4.2 and $\|\operatorname{Re} f\|_\rho, \|\operatorname{Im} f\|_\rho \leq \|f\|_\rho$,

$$\frac{\|2\operatorname{Re}(jw^{j-1}\Gamma_1)\|_\rho}{\omega_j} \leq \frac{2j\rho^{j-1}\|\Gamma_1\|_\rho}{\alpha(j+2)\rho^{j+1}} \leq \frac{2\|\Gamma_1\|_\rho}{\alpha\rho^2}, \quad \frac{\|\operatorname{Im}(w^j\Gamma_0)\|_\rho}{\omega_j} \leq \frac{\|\Gamma_0\|_\rho}{\alpha\rho(j_*+2)}.$$

Finally $jw^{j-1}\bar{p}W^\circ = j(1 - |w|^2)^2w^{j-1-\Lambda}(w^\Lambda\bar{p}Q^\circ)$ with $j-1-\Lambda \geq 0$, and Lemma 8.3 with $N = j-1-\Lambda$ gives

$$\frac{\|2\operatorname{Re}(jw^{j-1}\bar{p}W^\circ)\|_\rho}{\omega_j} \leq \frac{2j}{\alpha(j+2)\rho^{j+1}} \sum_{m,t} |\varpi_{m,t}| \frac{4(2t+1)(2t+3)\rho^{m+j-1-\Lambda}}{(m+j-\Lambda)^2} \leq \mathbf{U}_W.$$

Adding the three contributions proves the claim. $\square$

Remark 11.15. This far-shape bound is simpler and cruder than Proposition 8.4: it does not exploit the cancellation between $D\mathcal{F}_6(x^\circ)e_j$ and T_j in the harmonic rows, and $\mathbf{U}_\Gamma$ does not tend to zero as $j_* \rightarrow \infty$. It suffices here because Γ_1 and Γ_0 measure the deviation of the centre from the disc, which is small compared with $\alpha = b^3/2 \approx 1.2 \cdot 10^4$; the directly computed normalised column at $j = 397$ is about 0.082, while $Z_{\text{sh},\infty} < 0.465$ (Table 2).

Collecting the five classes, the number

$$Z = \max\{Z_f, Z_{g,\partial}, Z_{g,\infty}, Z_{\text{sh},\partial}, Z_{\text{sh},\infty}\}$$

bounds $\|I - AD\mathcal{F}_6(x^\circ)\|_{X \rightarrow X}$.

11.8. Certified bounds and the exact zero. The quantities defined above are finite expressions in the 465 dyadic centre coefficients, the entries of $\hat{A}$ and $\rho = 11/10$. They were evaluated in ball arithmetic at 128 bits as described in Section 13.3. Table 2 lists the results, rounded outward.

Theorem 11.16. *Let $r = 1/5000$. There is exactly one $x^* = (g^*, c^*) \in X_6$ with $\|x^* - x^\circ\|_X \leq r$ and $\mathcal{F}_6(x^*) = 0$.*

Proof. The support condition of Proposition 11.14, $j_* - 1 - 170 = 230 > M$, holds, and (11.12) holds by Table 2; hence A is bounded and injective. By Table 2, Proposition 11.8 applies with the rational majorants $Y = 53 \cdot 10^{-6}$, $Z = 47/100$, $C_2 = 251/1000$, $C_3 = 3 \cdot 10^{-7}$ and $C_4 = 3 \cdot 10^{-15}$. For these values and $r = 1/5000$, exact rational arithmetic gives

$$Y + (Z - 1)r + C_2r^2 + C_3r^3 + C_4r^4 = -5.29899599999975999999999952 \cdot 10^{-5} < 0,$$

$$Z - 1 + 2C_2r + 3C_3r^2 + 4C_4r^3 = -0.5298995999999639999999999904 < 0. \quad \square$$

Remark 11.17 (The centre for $n = 6$). The centre x° was obtained numerically, and its provenance plays no role in the proof. A Fourier–Bessel collocation in $\mathbb{R}^6$, with u expanded in the regular Helmholtz solutions $r^{-2}J_{\ell+2}(\rho_*r)U_{\ell/2}(\cos 2\phi)$, $\ell \equiv 0 \pmod{4}$, at the ball wavenumber $\rho_* = j_{3,8}$ and seeded with the second-order amplitude predicted by the mechanism of Section 14.2, produced a numerical solution of (1.2) with Cauchy defects of order 10^{-13} . Its

Quantity	Where defined	Certified bound
$b = c_1^\circ$	Section 11	28.864750643300045... (exact dyadic)
P, C	Lemma 11.7	$< 31.143514, < 32.088747$
V_0	Lemma 11.7	< 7841.683303
$\ I - \hat{A}M_f\ $	(11.12)	$< 6.714 \cdot 10^{-11}$
$\ R\ $	Lemma 11.12	< 1402.313800
H	Lemma 11.12	$< 5.848182 \cdot 10^8$
A_t ($N_H = 458$)	Lemma 11.12	< 3.164016
$\ A\ $	Lemma 11.12	< 1402.314
Y	Section 11	$< 5.243310 \cdot 10^{-5}$
Z_f (465 columns)	Section 11	< 0.073227
P_s, P_t	Lemma 11.13	$< 31.143246, < 2.675771 \cdot 10^{-4}$
$\delta_a = 2P_s P_t + P_t^2$	Lemma 11.13	< 0.016667
$Z_{g,\partial}$ (705 columns, max. at (62, 1))	Lemma 11.13	< 0.313291
$Z_{g,\infty}$	Lemma 11.13	< 0.390424
$Z_{\text{sh},\partial}$ (85 columns, max. at $j = 61$)	Section 11	< 0.387180
$\ \Gamma_1\ _\rho, \ \Gamma_0\ _\rho$	Proposition 11.14	$< 844.817923, < 4110.324378$
$\sum_{m,t} \varpi_{m,t} (2t+1)(2t+3)\rho^{m-\Lambda-2}$	Proposition 11.14	< 3721574.315769
$\mathbf{U}_\Gamma, \mathbf{U}_W$	Proposition 11.14	$< 0.11689839, < 0.03005935$
$Z_{\text{sh},\infty}$ (all $j \geq 401$)	Proposition 11.14	< 0.464977
Z	Section 11	< 0.464977
$C_2 = \ A\ c_2$	Lemma 11.7	< 0.250135
$C_3 = \ A\ c_3$	Lemma 11.7	$< 2.7601 \cdot 10^{-7}$
$C_4 = \ A\ c_4$	Lemma 11.7	$< 2.1035 \cdot 10^{-15}$

TABLE 2. Certified bounds for $n = 6$, rounded outward, for $M = 58$, $S = 30$, $\rho = 11/10$, $j_* = 401$. The complete interval enclosures are recorded in the certificate `certificate_r6.json` and in the seven interval receipts (Section 13.3); P, C and V_0 are intermediate values of the final verification script, which uses $\|A\| \leq 1402.314$ in C_2, C_3, C_4 .

meridian domain was mapped conformally to the disc by Theodorsen's method, and Newton's method was applied to the Galerkin truncation of (11.3) to the 15 shape coefficients $c_1, c_5, \dots, c_{57}$ and the 450 coefficients of g described above, with residual about 10^{-10} in double precision. The resulting binary64 numbers were then frozen and are interpreted as exact dyadic rationals. (The wavenumber ρ_* used to locate the centre is unrelated to the weight $\rho = 11/10$ of the proof.)

11.9. Reconstruction, geometry and convexity. Let $x^* = (g^*, c^*)$ be the zero of Theorem 11.16, $\psi^* = \psi_{c^*}$ and $\delta c = c^* - c^\circ$. Then $\sum_j \omega_j |\delta c_j| \leq r$, and consequently

$$(11.17) \quad |c_1^* - b| \leq \frac{r}{\omega_1}, \quad \sum_j j \rho^{j-1} |\delta c_j| \leq \theta_1 r, \quad \sum_j \rho^j |\delta c_j| \leq \theta_0 r.$$

Proposition 11.18. *The map ψ^* is injective on the disc $\{|w| < 21/20\}$, and $\psi^{*'} has no zeros there. The domain $D = \psi^*(\mathbb{D})$ is bounded by the real-analytic Jordan curve $\psi^*(\partial\mathbb{D})$, it is strictly star-shaped with respect to 0, invariant under $z \mapsto \bar{z}$, $z \mapsto -\bar{z}$ and $z \mapsto i\bar{z}$, and it is not a disc.$*

Proof. Let $R = 21/20 < \rho$, $b_- = b - r/\omega_1 \leq c_1^*$, and

$$s_1 = \sum_{j \geq 5} j |c_j^\circ| R^{j-1} + \theta_1 r, \quad s_0 = \sum_{j \geq 5} |c_j^\circ| + \theta_0 r.$$

By (11.17) and $R < \rho$, $\sum_{j \geq 5} j |c_j^*| R^{j-1} \leq s_1$ and $\sum_{j \geq 5} |c_j^*| \leq s_0$. The computation (Section 13.3) certifies

$$(11.18) \quad b_- - s_1 > 27.2138856, \quad \frac{s_1 + s_0}{b_- - s_0} < 0.0634442, \quad |c_5^*| \geq |c_5^\circ| - \frac{r}{\omega_5} > 0.08529699.$$

Thus $\operatorname{Re} \psi'(w) \geq c_1^* - \sum_{j \geq 5} j |c_j^*| R^{j-1} > 27.21$ for $|w| \leq R$, which gives univalence on the convex disc $|w| < R$ exactly as in Proposition 10.1. For $|w| = 1$ we have $|\psi^*(w) - c_1^* w| \leq s_0$, $|w \psi^{*'}(w) - \psi^*(w)| \leq s_1$ and $|\psi^*(w)| \geq b_- - s_0 > 0$, hence $|w \psi^{*'} / \psi^* - 1| < 0.0635$; the argument of Proposition 10.1 shows that D is strictly star-shaped and that ∂D is a radial graph with positive real-analytic radius. The symmetries were shown at the beginning of this section. If D were a disc, its symmetries would force its centre to be 0, and the Schwarz lemma argument of Proposition 10.1 would show that ψ^* is linear, contradicting $c_5^* \neq 0$. $\square$

Proposition 11.19. *Let c be any real sequence indexed by $j \equiv 1 \pmod{4}$ with $\sum_j \omega_j |c_j - c_j^\circ| \leq r$, for instance $c = c^*$. Then for $|w| \leq 1$,*

$$|\psi'_c(w)| > 27.67533225, \quad |\psi''_c(w)| < 7.86831783, \quad \operatorname{Re} \left(1 + \frac{w \psi''_c(w)}{\psi'_c(w)} \right) > 0.71569200.$$

Consequently $D = \psi^*(\mathbb{D})$ is strictly convex, and $\Omega = \Pi_6^{-1}(D)$ is convex.

Proof. Let $A_1 = \sum_{j \geq 5} j |c_j^\circ|$, $A_2 = \sum_{j \geq 5} j(j-1) |c_j^\circ|$, $\delta = c - c^\circ$, and

$$E_1 = r \max \left\{ \frac{1}{\omega_1}, \frac{1}{\alpha \rho^6} \right\}, \quad E_2 = \frac{r}{\alpha \rho e \log \rho}.$$

We have $|\delta_1| + \sum_{j \geq 5} j |\delta_j| = \sum_j \kappa_j \omega_j |\delta_j| \leq r \sup_j \kappa_j$ with $\kappa_1 = 1/\omega_1$ and $\kappa_j = j/\omega_j = j/((j+2)\alpha\rho^{j+1}) < 1/(\alpha\rho^6)$ for $j \geq 5$; hence this sum is at most E_1 . For $j \geq 5$, $j(j-1)/\omega_j < j\rho^{-j}/(\alpha\rho) \leq 1/(\alpha\rho e \log \rho)$, since $x \mapsto x\rho^{-x}$ attains its maximum $1/(e \log \rho)$ on $(0, \infty)$ at $x = 1/\log \rho$; hence $\sum_{j \geq 5} j(j-1) |\delta_j| \leq E_2$. Both suprema are over all j , so the infinite tail of c is covered. For $|w| \leq 1$ it follows that

$$|\psi'_c(w)| \geq c_1 - \sum_{j \geq 5} j |c_j| \geq b - A_1 - E_1 =: L, \quad |\psi''_c(w)| \leq \sum_{j \geq 5} j(j-1) |c_j| \leq A_2 + E_2 =: U.$$

The computation certifies $A_1 < 1.18941839$, $A_2 < 7.86831777$, $E_1 < 9.39 \cdot 10^{-9}$ and $E_2 < 5.84 \cdot 10^{-8}$, hence $L > 27.67533225$, $U < 7.86831783$ and $\operatorname{Re}(1 + w \psi''_c / \psi'_c) \geq 1 - U/L > 0.71569200$.

For $c = c^*$, the argument of Proposition 10.2 (the analytic convexity criterion) shows that D is strictly convex. Now let $x = (x', x'')$, $z = (z', z'') \in \Omega$ and $\sigma \in [0, 1]$, and put $m = (m_1, m_2) = \sigma \Pi_6(x) + (1 - \sigma) \Pi_6(z)$. By convexity $m \in D$, and $m_1, m_2 \geq 0$. Since D is invariant under both reflections, it contains the four points $(\pm m_1, \pm m_2)$, and by convexity their convex hull, the rectangle $[-m_1, m_1] \times [-m_2, m_2]$. By the triangle inequality in each factor $\mathbb{R}^3$, $|\sigma x' + (1 - \sigma) z'| \leq m_1$ and $|\sigma x'' + (1 - \sigma) z''| \leq m_2$, so $\Pi_6(\sigma x + (1 - \sigma) z) \in D$, that is, $\sigma x + (1 - \sigma) z \in \Omega$. $\square$

11.10. Proof of Theorem 1.1 for $n = 6$. Let $V^* = Kg^* + h_{c^*}$. By Lemma 11.6, $V^* \in C^1(\overline{\mathbb{D}})$ satisfies (11.2) with $\psi = \psi^*$ and has the parities required in Lemma 11.4. By Proposition 11.18, ψ^* is injective with nonvanishing derivative on a neighbourhood of $\overline{\mathbb{D}}$. Lemma 11.4 shows that $F = V^* \circ (\psi^*)^{-1} \in C^1(\overline{D})$ is a weak solution of $\Delta F + F = 0$ in D with $F = y_1 y_2$ and $\nabla F = (y_2, y_1)$ on ∂D , odd in y_1 and in y_2 , and invariant under the exchange of y_1 and y_2 . By Lemma 10.3 and Lemma 10.4 in the form of Lemma 11.5(f), F extends to a real-analytic solution on an open neighbourhood of $\overline{D}$; replacing this neighbourhood by the connected component containing $\overline{D}$ of its intersection with its images under the two reflections and the exchange, we obtain a neighbourhood N with these symmetries on which, by the identity theorem, F keeps its parities.

Since D is strictly star-shaped and symmetric, ∂D meets each axis in exactly two points. Proposition 11.3 shows that $\Omega = \Pi_6^{-1}(D)$ and $u = F(|x'|, |x''|)/(|x'| |x''|)$, extended analytically to $\{x' = 0\} \cup \{x'' = 0\}$, satisfy (1.2), that u is real analytic on the neighbourhood $\Pi_6^{-1}(N)$ of $\overline{\Omega}$, and that u is nonconstant.

Geometry of Ω . Let $R_D(\phi)$ be the radial function of D . It is positive and real analytic, even (by $z \mapsto \bar{z}$) and invariant under $\phi \mapsto \pi - \phi$ (by $z \mapsto -\bar{z}$), hence π -periodic, and its Fourier series has the form $R_D(\phi) = \sum_{k \geq 0} \alpha_k \cos 2k\phi$, with exponentially decaying coefficients (by the exchange symmetry $\alpha_k = 0$ for odd k , which we do not need). As in the case $n = 4$, $\tilde{R}(\tau) = \sum_k \alpha_k T_k(\tau)$ converges on a complex neighbourhood of $[-1, 1]$ and defines a real-analytic function with $\tilde{R}(\cos 2\phi) = R_D(\phi)$, positive near $[-1, 1]$. For $x \neq 0$ we have $\Pi_6(x) = |x|(\cos \phi, \sin \phi)$ with $\phi \in [0, \pi/2]$ and $\cos 2\phi = (|x'|^2 - |x''|^2)/|x|^2$. Hence

$$\Omega = \{0\} \cup \left\{ x \neq 0 : |x| < \tilde{R}\left(\frac{|x'|^2 - |x''|^2}{|x|^2}\right) \right\},$$

$$\partial\Omega = \left\{ \tilde{R}(|\omega'|^2 - |\omega''|^2) \omega : \omega = (\omega', \omega'') \in S^5 \right\}.$$

Thus Ω is strictly star-shaped with respect to 0, in particular connected. The map $\omega \mapsto \tilde{R}(|\omega'|^2 - |\omega''|^2) \omega$ is real analytic and injective on S^5 , and, exactly as for $n = 4$, its differential maps a tangent vector $v \perp \omega$ to a vector whose component tangent to S^5 is $\tilde{R}(|\omega'|^2 - |\omega''|^2) v \neq 0$. Hence $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^5 . The symmetries of Ω follow from $\Omega = \Pi_6^{-1}(D)$ and the invariance of D under the exchange of coordinates.

Ω is not a ball. If it were, its centre would be fixed by every isometry of $\mathbb{R}^6$ preserving Ω , in particular by $O(3) \times O(3)$, whose only fixed point is 0. Then $\Pi_6(\Omega) = D \cap [0, \infty)^2$ would be a quarter of a disc centred at 0, and by the reflection symmetries D would be that disc, contradicting Proposition 11.18. This completes the proof of Theorem 1.1 for $n = 6$, and Proposition 11.19 proves Corollary 1.3 for $n = 6$. $\square$

12. THE POMPEIU PROPERTY

Proof of Corollary 1.2. Let $n \in \{4, 6\}$, let Ω and u be as in Theorem 1.1 (proved in Sections 10 and 11), let $\xi \in \mathbb{R}^n$ with $|\xi| = 1$, and let $v(x) = e^{i\xi \cdot x}$, so that $\Delta v + v = 0$. Since u and v are smooth up to the real-analytic boundary and solve the same equation, Green's identity and (1.2) give

$$0 = \int_{\Omega} (u \Delta v - v \Delta u) dx = \int_{\partial\Omega} (u \partial_\nu v - v \partial_\nu u) dS = \int_{\partial\Omega} \partial_\nu v dS = \int_{\Omega} \Delta v dx = - \int_{\Omega} e^{i\xi \cdot x} dx.$$

Thus $\widehat{\mathbf{1}}_\Omega$ vanishes on the unit sphere. If $\sigma(x) = Qx + a$ is a rigid motion, then

$$\int_{\sigma(\Omega)} e^{ix_1} dx = e^{ia_1} \int_{\Omega} e^{i(Q^T e_1) \cdot x} dx = 0,$$

and taking real parts, $\int_{\sigma(\Omega)} \cos x_1 dx = 0$. Since $\cos x_1$ is continuous and not identically zero, Ω fails the Pompeiu property. By Theorem 1.1 and Corollary 1.3, Ω is a bounded convex domain, not a ball, with real-analytic boundary diffeomorphic to S^{n-1} , so the Pompeiu conjecture fails in $\mathbb{R}^n$. $\square$

13. COMPUTER-ASSISTED VERIFICATION

13.1. Dimension four: what the script checks. All finite computations for $n = 4$ are carried out by a single script, `verify_r4.py` (569 lines), which uses only the Python standard library and `python-flint` 0.9.0 [20], the Python interface to the ball arithmetic of FLINT/Arb [25]. Every real number is represented by a ball $[m \pm \varepsilon]$ whose arithmetic is rigorous, including all rounding errors. The 525 centre coefficients are embedded in the script as hexadecimal binary64 strings, which are converted to exact dyadic rationals (exactness is checked); $\rho = 21/20$, $r = 1/50000$ and the rational majorants of Theorem 9.1 are exact. The script performs the following steps.

- (1) It expands $\mathcal{F}(x^\circ)$ and the 525 finite columns of $D\mathcal{F}(x^\circ)$ in ball arithmetic, using (4.3), (5.1) and (6.3); no coefficient is discarded.
- (2) It assembles M_f , computes a ball enclosure of M_f^{-1} , and defines $\hat{A}$ as the matrix of the (exact, dyadic) midpoints of this enclosure. It then computes $I - \hat{A}M_f$ in ball arithmetic and verifies (7.5) for the weighted column norm. The matrix $\hat{A}$ is never treated as an interval-valued operator.
- (3) It computes $\|R\|$, H , the numbers h_n for $M + 2 \leq n \leq N_H = 443$ and the remainder bound of Lemma 7.6, hence A_t and a bound for $\|A\|$.
- (4) It computes Y and Z_f by applying A , in the form (7.6), (7.3), (7.4), to the residual and to the finite columns.
- (5) It computes the 3650 tail g -columns with $n \leq 123$, $s \leq 67$ with the short polynomial $p_{\leq 16}$ and adds the error term of (8.1); it evaluates $Z_{g,\infty}$.
- (6) It computes the 480 shape columns $43 \leq j \leq 1001$ without truncation, checks the hypotheses of Proposition 8.4, and evaluates $U_1, \dots, U_5, U_W$ and $Z_{\text{sh},\infty}$.
- (7) It evaluates Z , c_2 , c_3 , C_2 , C_3 , checks $Y < 7 \cdot 10^{-6}$, $Z < 37/100$, $C_2 < 4$, $C_3 < 10^{-3}$, and then the two inequalities (7.1) at $r = 1/50000$.
- (8) It checks the geometric inequalities (10.2) and (10.3).

In `python-flint`, an order relation between balls evaluates to true only if it holds for all points of the balls. Every check is an explicit test that raises an exception when it fails (also under `python -O`); maxima over columns are taken over rigorous upper endpoints, and nonfinite enclosures are rejected. After step (4) the script writes an intermediate checkpoint file (`*.base.json`), which carries no status field; only when all checks succeed does it write the certificate file (JSON) with the enclosures of all quantities of Table 1 and the status `PROVED`, and print `PROVED`. A diagnostic mode (`--stage base`) stops after step (4) and never prints `PROVED`.

A second script, `verify_convex.py`, checks that the centre data file coincides with the data embedded in `verify_r4.py`, that the main certificate has status `PROVED` for the radius $r = 1/50000$, and then verifies the bounds of Proposition 10.2, which involve only the finite sums A_1, A_2 and the explicit tail constants E_1, E_2 .

13.2. Dimension four: reproduction. From a directory containing the files `verify_r4.py`, `verify_convex.py` and `data/centre.json`, the certificate is reproduced by

```
mkdir -p logs
```

```
nice -n 19 env OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 \
  MKL_NUM_THREADS=1 python3 verify_r4.py --stage all --bits 128 --jstar 1001 \
  --output logs/certificate_r4_128.json
python3 verify_convex.py --bits 128 \
  --main-certificate logs/certificate_r4_128.json \
  --output logs/convex_128.json
```

The second command prints `PROVED` as its last line, the third prints `PROVED_CONVEX`. The runs use one thread. The distributed certificates are in the subdirectory `certificates/` (Section 13.4); the files written to `logs/` are byte-identical to them and have the checksums of Table 3. On a single core of a Linux server the 128-bit run took between 163 and 167 seconds of wall-clock time in several independent runs, and the same run at 192 bits (`--bits 192`) took about 198 seconds; the convexity check takes a fraction of a second. Runs under Python 3.10.19 and Python 3.12.3, both with `python-flint` 0.9.0, produced byte-identical certificate files. At 192 bits all inequalities hold with the same displayed bounds, and the finite inverse defect (7.5) is below $7.5 \cdot 10^{-54}$.

File	SHA-256
<code>verify_r4.py</code>	32c4334a215df722651b610f3c2922b3472b65edab2be21d32df7739195f85ed
<code>verify_convex.py</code>	ccb0ed709443dae7911a75ad3e6b3d355a2ea5320cf944ce53615d0cef14236
<code>data/centre.json</code>	ad87d8832320b1a94e1aadb04dfe4a523b9edf94941cbb00e219317b1bdb8fbc0
<code>certificate_r4_128.json</code>	59bba629550c19cedaa406f0c05fa8951ec40fbb2a954665b187367184a648bb
<code>certificate_r4_192.json</code>	068f73865cbb191d5574ef5abbbbf87774d49817af2f07f5daf3b6b1e1c20f73
<code>convex_128.json</code>	f14415b400133c47579baf08e8fda4e479826d55bc4c4d57a4dad134124f672e
<code>convex_192.json</code>	073794c76c79e6f7ae74c68f23af43636ab91a6990bde0ed5e59ba21475bb09f

TABLE 3. SHA-256 checksums of the verification scripts, the centre data and the certificates for $n = 4$.

13.3. Dimension six. For $n = 6$ the finite computations are split into five stage scripts, which evaluate the quantities of Section 11 in ball arithmetic at 128 bits, in batches, and write interval receipts (JSON files), and a final script that combines the receipts. All scripts use Python, `python-flint` 0.9.0 and, only to read the matrix $\hat{A}$, NumPy.

- (1) `finite_interval_r6.py` expands $\mathcal{F}_6(x^\circ)$ without truncation and computes $A\mathcal{F}_6(x^\circ)$ and Y (receipt `residual_bound_r6.json`); in eight batches covering all 465 finite columns it computes the inverse defect (11.12) and Z_f (`finite_batches_r6.json`). The matrix $\hat{A}$ is read from `inverse_r6_M58_S30.npy`, and its binary64 entries are converted to exact dyadic rationals.
- (2) `tail_inverse_r6.py` computes H , the numbers h_n for $M + 4 \leq n \leq N_H = 458$ and the remainder bound of Lemma 11.12, hence A_t (`tail_inverse_bound_r6.json`).
- (3) `g_tail_interval_r6.py` computes the 705 near tail g -columns in seven batches with the short polynomial $p_{\leq 24}$ and adds the error term of (11.15) (`g_tail_batches_r6.json`); it also evaluates the two far bounds of Lemma 11.13 (`g_far_bound_r6.json`).
- (4) `shape_tail_interval_r6.py` computes the 85 near shape columns $61 \leq j \leq 397$ without truncation, in five batches (`shape_tail_batches_r6.json`).
- (5) `far_shape_r6.py` computes Γ_1, Γ_0 , the expansion of Q° by the connection recurrences, the coefficients $\varpi_{m,t}$, U_Γ , U_W and $Z_{\text{sh},\infty}$ for $j_* = 401$, and checks the support condition of Proposition 11.14 (`far_shape_bound_r6.json`).

- (6) `verify_r6.py --stage final` checks the SHA-256 checksums of the centre, of $\hat{A}$, of the seven receipts and of the five stage scripts against values frozen in its source; checks that the batches of each receipt are contiguous and cover all columns of their class; recomputes $\|R\|$ from $\hat{A}$; forms Z and the constants c_2, c_3, c_4 of Lemma 11.7 with $\|A\| \leq 1402.314$; checks the rational majorants of Theorem 11.16 and the inequalities (11.6) at $r = 1/5000$; and checks the geometric inequalities (11.18). Only then does it write the certificate `certificate_r6.json`, with the status `PROVED`, all enclosures, the frozen checksums and its own checksum, and print `PROVED`.
- (7) `verify_convex_r6.py` checks the checksums of `verify_r6.py` and of the centre, and the status, radius, weight and checksums recorded in the certificate, and then verifies the bounds of Proposition 11.19 (`convex_r6.json`, status `PROVED_CONVEX`).

As for $n = 4$, an order relation between balls is true only if it holds for all points of the balls, every check raises an exception when it fails, and maxima are taken over rigorous upper endpoints; a NaN or infinite enclosure, whether computed or read back from a receipt, stops the verification. The script `test_fail_closed_r6.py` tests this behaviour on artificial inputs. A further script, `check_identities_r6.py`, checks in exact rational arithmetic the identities of Lemma 5.1(a)–(d) for all $n \equiv 2 \pmod{4}$, $n \leq 202$, and $1 \leq s \leq 80$ (4080 pairs), reruns the connection recurrences used for Q° in rational arithmetic (450 pairs, all coefficients positive), checks $\|Kg^\circ - (1 - |w|^2)^2 Q^\circ\|_\rho \leq 2.02 \cdot 10^{-31}$ in ball arithmetic, and compares (11.8) with the column algebra of the verifier, exactly for $5 \leq j \leq 2001$ and in ball arithmetic for $5 \leq j \leq 1001$. These identities are proved in Section 11; the script is a consistency check.

The complete six-dimensional verification is reproduced, from a directory containing the files of Table 4, by

```
sha256sum -c SHA256SUMS
bash reproduce_r6.sh
```

The script `reproduce_r6.sh` uses the interpreter given by the environment variable `PYTHON`, if it is set, checks the `python-flint` version, runs `test_fail_closed_r6.py`, regenerates all seven receipts from the centre and $\hat{A}$ with the stage scripts, each batch in a separate single-core process under `nice -n 19` with a time limit of 600 seconds, and compares them byte for byte with the distributed receipts. It then reruns steps (6) and (7) and `check_identities_r6.py` in a scratch directory, compares the resulting `certificate_r6.json`, `convex_r6.json` and `identities_r6.json` byte for byte with the distributed files, and prints `REPRODUCTION COMPLETE`; any mismatch stops it with a nonzero exit code. On one core of a Linux server, with Python 3.10.19 and `python-flint` 0.9.0, the complete reproduction took about three minutes of wall-clock time (3 min 2 s and 3 min 5 s in two independent runs). The individual final steps are

```
python3 verify_r6.py --stage final \
  --output certificate_r6.json
python3 verify_convex_r6.py
python3 check_identities_r6.py --output identities_r6.json
```

which print `PROVED`, `PROVED_CONVEX` and `IDENTITIES VERIFIED` and rewrite the three files byte-identically; they take about two seconds, less than a second and about fifteen seconds, respectively.

The analytic content of the proofs, namely all lemmas and all tail estimates, is contained in Sections 2–12; the scripts only evaluate the finitely many explicit expressions defined there.

13.4. Availability. The files listed in Tables 3 and 4 are distributed as ancillary files with the arXiv version of this paper: the four-dimensional files in the directory `anc/`, with the four

File	SHA-256
verify_r6.py	67a565438b6a7e459145a18b40c8c12d21134b097280575fce4bc9db8b6941c9
verify_convex_r6.py	be964ff20884b78025b2e09629639164bf8a35ac054750eec5924a127d457fe
check_identities_r6.py	23ff08666c23be4002b1011e4c07657eede0616660fae099a4d6ebad0ccb3d6c
test_fail_closed_r6.py	cf9f229636ddccbd9744a62098de0804791d396e1f585e11c7976a1f41478112
reproduce_r6.sh	c5fb06ca32d2058ae3862cbdc75e5b313185aae8559ad86d3315e2f514ebfd03
finite_interval_r6.py	dec54274aa7b52940ef6cf9253102b5d12393629b2bd40434e2b21a526dc060c
tail_inverse_r6.py	ea14f859d44a9f643263b73b10a4f95548c7b603cf6e7e3c3b6e109164c183ab
g_tail_interval_r6.py	6799e59083a354db0178e06dcbbc03f468f8f1edfdc12978fb85a3e2a0dca211
shape_tail_interval_r6.py	05566777c507e8c37ecc5e3fd20cbfda2010cb1b8a1d8825bede55a77ecbbb3c
far_shape_r6.py	99f1da9274008d3122bb5e7fffbdf949465f9d530639935c548674a005fe2804
center_r6.json	83018f0975e0cd61f4168e2de8bb1946032265869b5bfa168773c5920198aaba
inverse_r6_M58_S30.npy	8bb1b34aa882e1dbbe6943cba4726ba4eb18aefa4150cb734665a85b9f1026a9
finite_batches_r6.json	138e166004e655a3e76f3ceea5e6d48fc99549734d6de28f573246176044c97c
residual_bound_r6.json	b111bf5b26279810d582125f0d04fa07f07614d3e0bb60cf0119049319db556a
tail_inverse_bound_r6.json	393643e94df5e6d32189911ae61c10664e3658e6fff06d961931ef7d07f0433a
g_tail_batches_r6.json	194efaeb46b2762d86532cc9b91d6c6b652eef1ead0a879ceb082ed3cdd98375
g_far_bound_r6.json	8af49da5adc3aa3801c898949ea4ab8ff55c7ffa0ce94b51cd601b4ba300e474
shape_tail_batches_r6.json	7331746c80180533ce527565064aa39749ceca22172ae407f5f9d6a3531ce8ef
far_shape_bound_r6.json	6cf4938b1aad6a1f40f7acc11ae3e602b615b6a9d469cb319deff6595153478
certificate_r6.json	e481cb505c3d35219540f49153e828a9d94d3ebfb97ce2f657e055cba1daf66d
convex_r6.json	a7d42cd53f9a840096accf2bffb22cc0c8bd4a7ddde7fb42b0640d1299927f6
identities_r6.json	26540b55e9b07ce340ca6ebfc037e7ba8fb0497c49a82760c5460b18830169ca
SHA256SUMS	c05aa958160ff62136bc9053d6bfff223f765a259e2f6a827340bee3a86358e17

TABLE 4. SHA-256 checksums of the scripts, data, interval receipts and certificates for $n = 6$. The file `SHA256SUMS` lists these files (except itself) together with the two documentation files `FINAL.md` and `PROOF_ADDENDUM.md`.

certificates in `anc/certificates/`, and the six-dimensional files, together with `SHA256SUMS`, a file `RELEASE_README.md` with instructions, and the working notes `PROOF_ADDENDUM.md` and `FINAL.md` (the latter in Chinese), in the directory `anc/r6/`. All of these files are also available at <https://github.com/aster2024/schiffer-pompeiu-r4> (four-dimensional files in `verification/`, six-dimensional files in `verification/r6/`) and archived on Zenodo, <https://doi.org/10.5281/zenodo.22999505>.

13.5. Consistency checks. The following checks are not part of the proof. The algebraic identities of Sections 4, 5 and Lemma 7.3 were verified symbolically for small indices. An independently written floating-point implementation, which computes projections by quadrature instead of the recurrences (4.3) and uses its own approximate inverse, reproduced $\|R\|$, H , Z_f and the largest shape column of Table 1 to the displayed digits. The function u was also evaluated directly in $\mathbb{R}^4$ from a Fourier–Bessel representation $F = \sum_{m \text{ odd}} a_m J_m(r) \sin m\phi$ fitted to the boundary data of the centre: at random points of $\partial\Omega$ the defects $|u - 1|$ and $|\nabla u|$ were below 10^{-10} , and the Fourier transform of $\mathbf{1}_\Omega$, computed by quadrature, was below 10^{-10} in modulus on the unit sphere and between 18 and 28 in modulus on the sphere of radius 0.999.

For $n = 6$, Lemma 11.1 and the averages $\langle Y_\ell^3 \rangle / \langle Y_\ell^2 \rangle$ of Section 14.2, for $\ell = 4, 8, 12, 16$, were verified symbolically, and seven exact finite columns of $D\mathcal{F}_6(x^\circ)$ agreed with an independently written quadrature implementation to relative accuracy $1.7 \cdot 10^{-13}$. The Fourier–Bessel solution from which the centre was obtained (Remark 11.17) has Cauchy defects $8.2 \cdot 10^{-14}$ and $2.2 \cdot 10^{-13}$ at 2001 boundary points off the collocation grid, with 20 modes, and a solution with 28 modes agrees with it. Using only its boundary, the first 14 volume integrals over Ω of $O(3) \times O(3)$ -invariant regular solutions of $\Delta v + v = 0$, which vanish when $\widehat{\mathbf{1}}_\Omega$ vanishes on the unit sphere, were computed by quadrature; relative to the corresponding integrals in which the angular

integrand is replaced by its modulus, they were between 0 and $3.3 \cdot 10^{-13}$, compared with $4.7 \cdot 10^{-3}$ for a comparison domain that retains only the dominant degree-eight term of the boundary.

14. REMARKS ON OTHER DIMENSIONS AND ON CURVED SPACES

The statements in this section are not used in the proof of Theorem 1.1. Those about dimensions three and five are based on numerical computations only and are *not proved*, and Remark 14.1 is an observation that we have not pursued.

14.1. Symmetry classes for which the reduction is exact. Remark 2.5 shows that, among splittings $\mathbb{R}^n = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ with symmetry group $O(d_1) \times O(d_2)$, a reduction to the planar Laplacian without potential exists only for $n \in \{2, 4, 6\}$. Other cohomogeneity-two symmetry classes have the same property.

Remark 14.1 (Adjoint representations of rank two). Up to finite index and the additional reflections, the symmetry groups used in this paper are the adjoint groups of the compact Lie algebras $\mathfrak{u}(2) \cong \mathbb{R}^4$ and $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2) \cong \mathbb{R}^6$, which have rank two, and the weights y and $y_1 y_2$ are, up to constants, the products δ of the positive roots on a Cartan subalgebra $\mathfrak{t}$. For a compact Lie algebra $\mathfrak{g}$, a classical formula of Harish-Chandra [22] (see also [23, Ch. II]) states that $(\Delta_{\mathfrak{g}} u)|_{\mathfrak{t}} = \delta^{-1} \Delta_{\mathfrak{t}}(\delta u|_{\mathfrak{t}})$ on the regular set for every Ad-invariant C^2 function u ; the polynomial δ is harmonic, being the anti-invariant of lowest degree of the Weyl group. For $\mathfrak{g}$ of rank two this gives a reduction of (1.1), for Ad-invariant u , to a planar Helmholtz problem with Cauchy data $(\delta, \nabla \delta)$ and without potential, also for $\mathfrak{su}(3) \cong \mathbb{R}^8$, $\mathfrak{sp}(2) \cong \mathfrak{so}(5) \cong \mathbb{R}^{10}$ and $\mathfrak{g}_2 \cong \mathbb{R}^{14}$, where δ is a harmonic polynomial of degree 3, 4 and 6, respectively. We have not examined whether the other steps of our argument, in particular the analytic reconstruction across the walls of the Weyl chambers and the existence of suitable near-resonances, carry over to these cases.

14.2. A transcritical mechanism. Let $n \geq 3$ and let $\rho_m = j_{n/2, m}$, so that the unit ball of $\mathbb{R}^n$ carries the radial solution of (1.1) with $\mu = \rho_m^2$. Consider domains $\{x : |x| < 1 + sY_{\ell}(x_1/|x|) + O(s^2)\}$ invariant under $O(n-1) \times \mathbb{Z}_2$, where Y_{ℓ} is the zonal spherical harmonic of even degree ℓ on S^{n-1} , normalised by $Y_{\ell}(1) = 1$. For $n = 6$ we use instead the class of Section 11, of domains invariant under $O(3) \times O(3)$ and the exchange of the factors; there the invariant spherical harmonics of degree $\ell = 2k$ are the functions $Y_{\ell}(\omega) = U_k(t)/(k+1)$ of $t = |\omega'|^2 - |\omega''|^2$, and the exchange symmetry requires k to be even. The linearisation of the Schiffer problem at the ball in the direction Y_{ℓ} is degenerate exactly when $J_{\ell+n/2-1}(\rho_m) = 0$, and this never happens for integer n and $\ell \geq 2$: for even n this is Bourget's hypothesis, proved by Siegel [36], that Bessel functions of distinct nonnegative integer orders have no common positive zero, and for odd n it follows from a similar transcendence argument based on the Lindemann–Weierstrass theorem. So there is no bifurcation from balls in this class. However, $J_{\ell+n/2-1}(\rho_m)$ can be small. A formal Lyapunov–Schmidt expansion in the direction Y_{ℓ} then gives, at second order, a reduced equation of the form

$$-\sigma s + \frac{\rho_m^2}{2} \frac{\langle Y_{\ell}^3 \rangle}{\langle Y_{\ell}^2 \rangle} s^2 + O(s^3, \sigma s^2) = 0,$$

where $\langle \cdot \rangle$ is the average over S^{n-1} and σ is a small quantity proportional to $J_{\ell+n/2-1}(\rho_m)$. This suggests a non-ball solution with boundary amplitude $s_* \approx 2\sigma \langle Y_{\ell}^2 \rangle / (\rho_m^2 \langle Y_{\ell}^3 \rangle)$.

In the plane the corresponding cubic average $\langle \cos^3 N\phi \rangle$ vanishes, the reduced equation is odd in s , and a sign condition or an additional parameter is needed; compare the relaxed

parameter of [7] and the continuation from the constant-flux problem in [11]. For $n \geq 3$ and even ℓ the cubic average does not vanish: for $n = 3$, $\langle P_\ell^3 \rangle / \langle P_\ell^2 \rangle = (2\ell+1) \begin{pmatrix} \ell & \ell & \ell \\ 0 & 0 & 0 \end{pmatrix}^2 > 0$; for $n = 4$ the zonal harmonics are $U_\ell(\cos \phi) / (\ell + 1)$, where $U_\ell(\cos \phi)$ is the character of the irreducible representation of $SU(2) \cong S^3$ of dimension $\ell + 1$, and the Clebsch–Gordan rule gives $\langle U_\ell^3 \rangle = \langle U_\ell^2 \rangle = 1$ for even ℓ . For $n = 6$ in the $O(3) \times O(3)$ -invariant class, the invariant measure of S^5 is carried by $t = |\omega'|^2 - |\omega''|^2$ to a multiple of $\sqrt{1-t^2} dt$, for which the U_k are orthonormal after normalisation, and $U_k^2 = \sum_{i=0}^k U_{2i}$ gives $\langle U_k^3 \rangle = \langle U_k^2 \rangle = 1$ for even k , hence $\langle Y_\ell^3 \rangle / \langle Y_\ell^2 \rangle = 1/(k+1)$. That quadratic terms make axisymmetric bifurcations in the even-degree representations of $O(3)$ generically transcritical is a classical fact of equivariant bifurcation theory [21, 24]. What we use is only the observation that this classical mechanism applies to the Schiffer problem near a near-resonance, where the planar obstruction disappears. Integrals of products of three spherical harmonics arise in any second-order perturbation analysis near a ball, and we make no claim of novelty for them.

For $(n, \rho_m, \ell) = (4, j_{2,4}, 6)$, the zero $j_{2,4} \approx 14.79595$ of J_2 is close to the zero $j_{7,2} \approx 14.82127$ of $J_7 = J_{\ell+n/2-1}$, and the formal prediction for the relative Y_6 -amplitude is about $2.4 \cdot 10^{-2}$. The domain of Theorem 1.1 has Y_6 -amplitude about $2.2 \cdot 10^{-2}$ (Remark 1.4); in the meridian plane the Y_6 direction corresponds to the mode $J_7(r) \sin 7\phi$ (Section 2). This is how the example was located. Numerically, Newton iterations started from the ball perturbed by a small multiple of Y_6 converge either back to the ball or to our domain, depending on the size of the perturbation, as expected from a transcritical picture.

For $(n, \rho_m, \ell) = (6, j_{3,8}, 8)$, the zero $j_{3,8} \approx 28.90835$ of J_3 is close to the zero $j_{10,5} \approx 28.88738$ of $J_{10} = J_{\ell+n/2-1}$, and $J_{10}(j_{3,8}) \approx -3.0 \cdot 10^{-3}$. The formal prediction for the relative amplitude of the degree-eight harmonic is about $7.3 \cdot 10^{-3}$, and the six-dimensional domain of Theorem 1.1 has amplitude about $7.2 \cdot 10^{-3}$ (Remark 1.4). In the meridian plane this direction corresponds to the mode $J_{10}(r) \sin 10\phi$ (Section 11). The centre of Section 11 was located in this way (Remark 11.17).

14.3. Dimensions three and five: numerical evidence. With a spectral collocation solver (expansions in exact Helmholtz solutions $r^{1-n/2} J_{\ell+n/2-1}(\sqrt{\mu} r) Y_\ell$, boundary given as a radial graph, Gauss–Newton iteration), we found numerical solutions of (1.1) on non-ball, star-shaped, $O(n-1) \times \mathbb{Z}_2$ -invariant domains near the near-resonances predicted in Section 14.2:

- $n = 3$, $\sqrt{\mu} = j_{3/2,42} \approx 133.5102$ (unit ball normalisation), $\ell = 20$, boundary amplitude about $9.6 \cdot 10^{-4}$. In multiprecision the collocation residual is about 10^{-36} , and the residual off the collocation grid decreases spectrally with the truncation (about 10^{-11} , 10^{-14} , 10^{-17} and 10^{-20} for 120, 160, 200 and 240 modes). A test of the Pompeiu condition that uses only the computed boundary gives $2 \cdot 10^{-12}$, compared with $1.4 \cdot 10^{-2}$ for a comparison domain.
- $n = 3$, $\sqrt{\mu} = j_{3/2,15} \approx 48.6741$, $\ell = 12$, amplitude about $1.5 \cdot 10^{-2}$, off-grid residual about $2 \cdot 10^{-11}$ with 150 modes.
- $n = 5$, $\sqrt{\mu} = j_{5/2,30} \approx 97.3586$, $\ell = 16$, amplitude about $7.1 \cdot 10^{-3}$, and $\sqrt{\mu} = j_{5/2,24} \approx 78.5016$, $\ell = 14$, amplitude about $2.1 \cdot 10^{-2}$; residuals about 10^{-11} , stable under changes of the truncation.

We do not claim that these solutions exist: they are not certified. By Remark 2.5, a reduction to the planar Laplacian of the type used in Sections 2 and 11 is not available in these dimensions, and a computer-assisted proof would require a different fixed-domain formulation.

14.4. Curved spaces. In spheres and hyperbolic spaces the curvature provides a genuine one-parameter family in which radial Neumann eigenvalues and non-radial Dirichlet eigenvalues of

geodesic balls can cross, which allows local bifurcation from geodesic balls. Cao-Labora and Fernández [8] used this to construct contractible Schiffer domains on the half-sphere S^2 , and they pointed out [8, Remark 1.2] that their method generalises to other non-flat spaces, with numerically observed crossings for S^3 , S^4 and H^2 . Earlier examples in spheres are bounded by isoparametric hypersurfaces [33, 35], whose boundaries are not topological spheres in general. The mechanism of Section 14.2 is also relevant in spheres and hyperbolic spaces of dimension $n \geq 3$; we do not pursue curved spaces in this paper.

15. RELATION WITH RIGIDITY RESULTS AND OTHER CLAIMS

Perturbations of balls. Several theorems assert rigidity near balls. Agranovsky [1] considers a real-analytic family of planar domains issuing from the disc along which an analytic branch of Pompeiu spectral parameters persists. Kobayashi [27] proves that balls are isolated among domains carrying such overdetermined eigenfunctions, for families of domains issuing from a ball, under a nondegeneracy condition and with uniformly bounded spectral parameters. Canuto [6] proves, in arbitrary dimension, rigidity near a fixed radial Neumann eigenvalue within an eigenvalue-dependent class of perturbations. These hypotheses concern solution-carrying families issuing from a ball, or specified perturbation classes, rather than the geometric proximity of a single domain to a ball, and where a neighbourhood of a ball is involved its size is not explicit. Theorem 1.1 concerns two single domains, whose boundaries deviate from the sphere of radius $j_{2,4}$ by up to 2.3% ($n = 4$) and from the sphere of radius $j_{3,8}$ by up to 0.74% ($n = 6$) (Remark 1.4), and there is no contradiction with these results. Heuristically, the size of such neighbourhoods is limited by the inverse of the linearised operator at the ball, which is large near a near-resonance, and the transcritical picture of Section 14.2 places the non-ball solution at a distance comparable to the near-resonance. Both examples sit at near-resonances, with small divisors $J_7(j_{2,4}) \approx -4.9 \cdot 10^{-3}$ and $J_{10}(j_{3,8}) \approx -3.0 \cdot 10^{-3}$. We have not attempted to make the constants in these theorems explicit.

Spectral conditions. A domain for which (1.1) is solvable for infinitely many μ is a ball [3, 4]; our construction provides one eigenvalue for each domain. Dai, Liu and Sun [12] prove a rigidity theorem in all dimensions when the eigenvalue in (1.1) is the second Dirichlet eigenvalue; for our domains, the eigenvalue 1 is much larger than the second Dirichlet eigenvalue of Ω . Indeed, for $n = 4$ the proof of Proposition 10.1 gives $|\psi^*| \geq b_- - s_0 > 14.54$ on $\partial\mathbb{D}$, so Ω contains the ball of radius 14.54 about the origin, and by domain monotonicity its second Dirichlet eigenvalue is at most $(j_{2,1}/14.54)^2 < 0.13$. For $n = 6$ the proof of Proposition 11.18 gives $|\psi^*| \geq b_- - s_0 > 28.69$ on $\partial\mathbb{D}$, and the second Dirichlet eigenvalue of Ω is at most $(j_{3,1}/28.69)^2 < 0.05$. In the plane, rigidity is known for low eigenvalues [15] and under further conditions [2, 14, 26, 29].

Convex domains and high frequency. Dai, Sun, Wei and Zhang [13] prove that a uniformly convex planar domain carrying solutions of (1.1) with sufficiently large eigenvalue, above a threshold depending on the domain, is a disc, and state that the extension to higher dimensions is left for future work. Their theorem is planar and does not apply to our examples. By Corollary 1.3, our domains are convex; thus convexity alone does not force Schiffer rigidity in $\mathbb{R}^4$ or $\mathbb{R}^6$ at every eigenvalue, and a higher-dimensional analogue of such high-frequency results could only hold above a frequency threshold that excludes the eigenvalues of our examples.

Claims in dimension three. Ramm [34] states that a bounded connected C^2 domain in $\mathbb{R}^3$ whose indicator function has Fourier transform vanishing on a sphere is a ball, and Chen [9] states a proof of the Schiffer conjecture for star-shaped Lipschitz domains in $\mathbb{R}^3$. Both concern $\mathbb{R}^3$, so they are not contradicted by Theorem 1.1, which concerns dimensions four and six. We note only that the numerical three-dimensional solutions of Section 14 are star-shaped

with analytic boundary, so that a rigorous version of them would be incompatible with these statements.

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